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Master Thesis in Theoretical Physics

NUMERICAL RELATIVITY STUDIES OF SPIN INDUCTION AND DYNAMICAL CAPTURE IN DENSE CLUSTERS OF BLACK HOLES

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Abstract

The first direct detection of gravitational waves by LIGO in 2015 marked a revolutionary breakthrough in our ability to study the Universe. Over the past decade, the LIGO-Virgo-KAGRA collaboration has identified over 80 gravitational-wave events, primarily originating from the mergers of compact objects such as black holes. These discoveries have significantly enhanced our understanding of astrophysical binaries, relativistic dynamics, and the nature of strong-field gravity.

Throughout this Thesis, I use Numerical Relativity simulations to investigate several relativistic phenomena that, while not yet observed by current interferometers, are expected to become increasingly relevant as detector sensitivity improves in the coming years. First, I will present previous studies on spin induction caused by hyperbolic encounters in dense clusters of black holes, examining how such interactions can influence the spin dynamics of compact objects. Additionally, I will explore the properties of dynamical captures — close hyperbolic encounters that can lead to an almost head-on collision after an effect of tidal capture. By analyzing these scenarios, this work aims to provide theoretical predictions that will help guide future gravitational-wave observations and deepen our understanding of relativistic astrophysics.

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I would like to also acknowledge that all the simulations shown throughout this Thesis have been run in the Hydra HPC cluster at the IFT.

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Introduction

A new Era started in 2015 with the first detection of a gravitational wave (GW) — in the event GW150914 [1] —; this new source to probe the Universe has forever revolutionized the field of astrophysics, astronomy and cosmology. With the ability to measure ripples in spacetime — caused by the acceleration of massive objects like black holes (BHs) or neutron stars (NSs) — we now have new powerful tools to study key events across the Universe. These range from probing the early Universe before the cosmic microwave background, through the stochastic gravitational wave background (SGWB), to detecting and analyzing black hole mergers and other strong-gravity events happening around us.

With the LIGO-Virgo-KAGRA (LVK) collaboration [2], as well as the pulsar timing arrays (PTAs, [4]); future ground-based detectors such as the Einstein Telescope (ET, [66]) and the Cosmic Explorer (CE, [70]); and future space-borne missions (LISA [9], DECIGO [50], etc.), our GW observations will cover an impressive range in the frequency band, helping us understand and detect a huge range of astrophysical phenomena. One promising possibility of these efforts is to detect primordial black holes (PBHs), which are a dark matter (DM) candidate that has gained attention over the last decade aside from Particle Physics. If these compact objects do exist, one of the most intriguing questions of Modern Physics will be answered.

However, extracting precise physical information from GW signals requires a solid theoretical foundation built on General Relativity (GR), advanced computational techniques, and an understanding of strong-field dynamics. Within GR, one of the foundational tools used to describe isolated gravitational systems is the Arnowitt-Deser-Misner (ADM) formalism. The ADM approach reformulates Einstein's equations as a Hamiltonian system, allowing the decomposition of spacetime into a three-dimensional spatial hypersurface evolving over time, which is a more natural way to describe a dynamical system compared to the covariant form of Einstein's Equations. This (3+1) formalism underpins many modern Numerical Relativity (NR) schemes, providing a clear separation between constraints and evolution equations.

Numerical Relativity bridges the gap between analytical and observational studies by offering a full nonlinear solution to Einstein's equations in dynamical regimes where no exact solutions exist. NR has been instrumental in modeling binary black hole mergers, neutron star collisions, and more recently, high-energy black hole scatterings. Simulations rely on sophisticated methods, such as finite differencing with Adaptive Mesh Refinement (AMR) and spectral techniques, along with stable formulations of Einstein's equations such as the Baumgarte-Shapiro-Shibata-Nakamura (BSSN) formalism (a modified version of the ADM formalism). By resolving the spacetime curvature in these extreme regimes, NR provides waveform templates, which are essential for GW data analysis and enables the study of phenomena such as spin induction, spin precession and tidal heating.

In Chapter 1, we will delve into the fundamental aspects of GW theory, ADM and BSSN formalisms to accurately understand how a NR simulation functions, as well as some indispensable properties of BHs and their interaction in a binary system.

This Thesis will thereafter investigate the dynamics of strong-field black hole interactions beyond the traditional quasicircular inspiral scenario, by focusing on close hyperbolic encounters (CHEs) and dynamical captures (DCs) — which also yield initial hyperbolic-like trajectories.

Unlike bound inspirals, hyperbolic interactions are characterized by small impact parameters, sub-ISCO¹ trajectories and brief interaction times. However, the nonlinear gravitational dynamics during such encounters can induce significant changes in the spin of each black hole, even without a merger. Our results indicate that spin-up mechanisms through scattering may contribute non-negligibly to the spin distribution of black holes observed by LIGO-Virgo-KAGRA.

Due to their novel gravitational wave signatures (with respect to bound inspirals), hyperbolic encounters have garnered significant attention over the last decade. Studies have ranged from foundational theoretical frameworks and analytical formalisms [38, 39, 78] to detailed modeling [20, 26, 34, 73], and investigations of their detectability with both current [16, 56] and future [14, 51, 57] GW observatories. The frequency and characteristics of these events are strongly dependent on the properties of the black hole populations involved. This is particularly relevant in the context of primordial black holes (PBHs) residing in dense clusters [27, 28, 36, 47, 61, 81], where such scattering events are expected to occur frequently. Understanding these interactions could provide critical insight into whether a portion of the black hole population is situated in such dense environments, as predicted by certain cosmological models, especially those concerning the thermal history of the early universe [21, 22]. Furthermore, if hyperbolic encounters occur with sufficient frequency, they could contribute to a stochastic gravitational wave background rich with information about black hole clustering [37].

In Chapter 2, we show some explicit details about the software we use to compute our simulations, and the structure of the initial conditions and general technical properties these simulations have. We also show the characteristic behaviors and evolutionary dynamics that the events we study throughout this Thesis follow, along with some key aspects of Hamiltonian theory to explain some fundamental characteristics of spin induction.

In Chapter 3, we will review a published work by myself and my supervisors (see [72]) on spin induction in close hyperbolic encounters; and in Chapter 4, we explore original work on dynamical capture. DCs are CHEs that lead to bound systems that end up producing a merger, after some time. We aim to study the properties of the emitted GWs in these events of capture, along with analyzing the evolution of the spins and masses of the initial and final BHs.

Therefore, this Thesis contributes to the growing body of work on eccentric and dynamical black hole interactions, expanding our understanding of gravitational wave sources beyond quasi-circular binaries. By combining theoretical insight, numerical simulation, and waveform analysis, we offer new perspectives on the spin and dynamical history of black holes formed through nonstandard channels. Future observational runs of GW detectors, combined with refined modeling of eccentric systems, will provide the opportunity to confirm these scenarios and distinguish them from the more familiar inspiral-merger-ringdown paradigm. As the sensitivity and bandwidth of detectors improve, the ability to identify and characterize such events will be critical for unraveling the complex dynamical processes that shape compact-object populations in the universe.

¹ISCO is the acronym for Innermost Stable Circular Orbit.

1. Theoretical background

For a better understanding of the achieved results in this project, it is necessary to introduce a few theoretical and numerical concepts that will appear throughout it. The main theoretical concepts discussed in the following are GWs and the dynamics of BHs. After that, since all of the results in this project were computed through simulations, we will need to explain some key features that our codes require to achieve reliable results. In this way, it will be possible to discuss the simulations with just the basics of GR and some physical intuition.

The primary references used for this chapter are the books written by Prof. Michele Maggiore [54, 55], which have already become the main textbooks for GW theory. Another reference, which was used for more technical aspects of this Thesis — regarding, more specifically, Numerical Relativity —, is the PhD Thesis of one of the supervisors of this work [48].

1.1. Gravitational wave formalism

A gravitational wave is a ripple in spacetime generated by sudden changes in the acceleration of massive objects. It represents spacetime’s dynamic response to the redistribution of energy and momentum. To fully understand the concept of GWs, we must describe them within the framework of Einstein’s field equations, which govern the interaction between matter and the geometry of spacetime. This set of equations is expressed as:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \frac{8\pi G}{c^4}T_{\mu\nu}, \quad (1.1.1)$$

where the Einstein tensor ($G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$) encodes information about the curvature of spacetime, and the stress-energy tensor $T_{\mu\nu}$ characterizes the matter and energy content.

For convenience, we adopt geometrized units ($G = c = 1$), which simplifies the equations and unifies time, distance, and mass into the same units, $[t] = [r] = [M]$. This is a standard approach in GR that greatly streamlines calculations.

The field equations tell us that matter and energy curve spacetime, and this curvature determines how objects move. When massive objects accelerate, this curvature changes dynamically, producing gravitational waves that propagate at the speed of light.

To study GWs, we consider the weak-field limit where spacetime is nearly flat, and the metric can be written as a small perturbation ($h_{\mu\nu}$) on the Minkowski background ($\eta_{\mu\nu}$):

$$g_{\mu\nu} \approx \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (1.1.2)$$

This approximation allows us to greatly simplify Einstein’s equations.

Applying an infinitesimal gauge (coordinate) transformation of the form:

$$x'^{\mu} = x^{\mu} - \xi^{\mu}(x), \quad (1.1.3)$$

the perturbation transforms as:

$$h'_{\mu\nu} = h_{\mu\nu} + \partial_\mu \xi_\nu + \partial_\nu \xi_\mu. \quad (1.1.4)$$

By computing the corresponding (linearized) Christoffel symbols and Riemann tensor:

$$\Gamma^\alpha_{\mu\nu} = \frac{1}{2} g^{\alpha\lambda} (\partial_\nu h_{\mu\lambda} + \partial_\mu h_{\nu\lambda} - \partial_\lambda h_{\mu\nu}), \quad (1.1.5)$$

$$R_{\alpha\beta\mu\nu} = \frac{1}{2} (\partial_\beta \partial_\mu h_{\alpha\nu} + \partial_\alpha \partial_\nu h_{\beta\mu} - \partial_\alpha \partial_\mu h_{\beta\nu} - \partial_\beta \partial_\nu h_{\alpha\mu}), \quad (1.1.6)$$

respectively; we derive the linearized wave equation:

$$\square \bar{h}_{\mu\nu} = -16\pi T_{\mu\nu}, \quad (1.1.7)$$

where $\square = -\partial_t^2 + \nabla^2$ is the d'Alembertian operator, and $\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} h \eta_{\mu\nu}$ is the trace-reversed metric perturbation, which fulfills the *harmonic gauge*:

$$\partial^\mu \bar{h}_{\mu\nu} = 0. \quad (1.1.8)$$

A more detailed derivation can be found in Maggiore's textbook on gravitational waves [54].

For the scenarios relevant to us — detectors located far from the GW source — we consider the vacuum solution where $T_{\mu\nu} = 0$. This simplifies the equation to:

$$\square h_{\mu\nu} = 0, \quad (1.1.9)$$

which is the standard wave equation describing waves traveling at the speed of light.

We then have freedom to choose a gauge (given by the ξ -covectors in Eq. (1.1.4)) that greatly simplifies the solutions. Firstly, we impose the *traceless* condition:

$$\bar{h} = 0, \quad (1.1.10)$$

so $\bar{h}_{\mu\nu} = h_{\mu\nu}$. Secondly, we use three additional conditions

$$h_{i0} = 0, \quad (1.1.11)$$

which are the *transversality* conditions. After this, and imposing the *harmonic gauge* (Eq. (1.1.8)), we can arrive to the condition

$$\partial^0 h_{00} = 0, \quad (1.1.12)$$

which means that h_{00} is constant in time, and since GWs describe the time-dependent part of the metric, we can set $h_{00} = 0$.

Therefore, we can write the conditions for this *traceless* and *transversal* gauge in the following manner:

$$h_{\mu 0} = 0, \quad h^i{}_i = 0, \quad \partial^i h_{ij} = 0. \quad (1.1.13)$$

Assuming the wave propagates along the z -axis, the solution in the Transverse-Traceless (TT) gauge is:

$$h_{\mu\nu}^{\text{TT}}(t, z) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cos \omega(t - z), \quad (1.1.14)$$

where h_+ and $h_\times$ represent the two independent GW polarizations. The TT gauge conditions ensure the wave is purely spatial, transverse to the direction of propagation, and traceless, as imposed by Eq. (1.1.13).

The relationship between the general spatial metric perturbation h_{ij} and its TT-projected form is given by:

$$h_{ij}^{\text{TT}} = \Lambda_{ij,kl} h_{kl}, \quad (1.1.15)$$

where the projection operator $\Lambda_{ij,kl}$ is defined as:

$$\Lambda_{ij,kl} = P_{ik}P_{jl} - \frac{1}{2}P_{ij}P_{kl}, \quad P_{ij} = \delta_{ij} - n_i n_j, \quad (1.1.16)$$

with $\vec{n}$ being the unit vector in the wave's propagation direction. This operator projects the perturbation onto the plane orthogonal to $\vec{n}$, ensuring transversality.

The key feature of GWs is their ability to stretch and compress spacetime as they pass through. Specifically, the h_+ component stretches space along one axis (say, y) while compressing it along the axis (x), orthogonal to the direction of propagation (z), and this behavior oscillates with the wave. The $h_\times$ polarization induces a similar effect but rotated by 45 degrees. This tidal distortion is what laser interferometers like LIGO and Virgo are designed to detect. The interferometers measure the differential stretching and compression of their orthogonal arms, directly tracking the time evolution of h_+ and $h_\times$ with extreme precision.

Similarly to electromagnetic waves, one of the relevant properties of GWs is that their amplitude decays as $\propto 1/r$ with distance from the source. Because of this, the strain (h) measured at a distance r^* can be rescaled to any other radius. To analyze the angular dependence of the wave, the strain is typically decomposed into spin-weighted spherical harmonics $Y_{-2}^{lm}(\theta, \phi)$, which naturally encode the quadrupolar nature of GWs:

$$h_{+, \times} = \sum_{l=2}^{\infty} \sum_{m=-l}^{+l} h_{+, \times}^{lm} Y_{-2}^{lm}(\theta, \phi). \quad (1.1.17)$$

Among these modes, the $(l = 2, m = 2)$ quadrupole mode dominates the GW signal in most astrophysical scenarios, especially for binary black hole (BBH) mergers. Higher-order modes such as $(3, 3)$ or $(4, 4)$ can become significant in cases involving asymmetries, high mass ratios, or strong precession, but the $(2, 2)$ mode remains the most prominent and is the primary focus of our analysis.

The Weyl scalar

The results in this project are obtained with Numerical Relativity (NR); however, the strain amplitude h is not possible to obtain as a direct output from numerical simulations. Hence, we must introduce a more fundamental quantity that describes the ripples in spacetime directly from the properties of the metric.

First, we define the Weyl tensor:

$$C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma} - \frac{1}{2}(g_{\mu\rho}R_{\nu\sigma} - g_{\mu\sigma}R_{\nu\rho} - g_{\nu\rho}R_{\mu\sigma} + g_{\nu\sigma}R_{\mu\rho}) + \frac{1}{6}R(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho}), \quad (1.1.18)$$

which is essentially the trace-free part of the Riemann tensor, meaning that it vanishes when contracting any pair of its indices. Additionally, the Weyl tensor possesses the same symmetries as the Riemann tensor.

Moreover, we want to project the components of the Weyl tensor onto a convenient basis to produce some scalars with physical significance. This is done in the Newman-Penrose formalism [55, 60] which introduces a *null tetrad*, which is a basis of four independent vectors $l^\mu, n^\mu, m^\mu, \bar{m}^\mu$ which have the following relations:

$$\begin{aligned} l^\mu l_\mu &= n^\mu n_\mu = m^\mu m_\mu = \bar{m}^\mu \bar{m}_\mu = 0, \\ m^\mu \bar{m}_\mu &= -l^\mu n_\mu = 1, \end{aligned} \quad (1.1.19)$$

from which follows that the metric can be written as

$$g^{\mu\nu} = m^\mu \bar{m}^\nu + \bar{m}^\mu m^\nu - l^\mu n^\nu - n^\mu l^\nu. \quad (1.1.20)$$

Using this null tetrad, we can compute the *Weyl scalars* which are defined in the following manner

$$\begin{aligned} \Psi_0 &= C_{\mu\nu\rho\sigma} l^\mu m^\nu l^\rho m^\sigma; \\ \Psi_1 &= C_{\mu\nu\rho\sigma} l^\mu n^\nu l^\rho m^\sigma; \\ \Psi_2 &= C_{\mu\nu\rho\sigma} l^\mu m^\nu \bar{m}^\rho n^\sigma; \\ \Psi_3 &= C_{\mu\nu\rho\sigma} n^\mu l^\nu n^\rho \bar{m}^\sigma; \\ \Psi_4 &= C_{\mu\nu\rho\sigma} n^\mu \bar{m}^\nu n^\rho \bar{m}^\sigma. \end{aligned} \quad (1.1.21)$$

Moreover, in vacuum, Einstein's equations yield $R_{\mu\nu} = 0$ which gives $C_{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma}$. The quantities Ψ_1, Ψ_2 and Ψ_3 are generally related to gravito-electric and gravito-magnetic fields and a Coulomb-like potential, and in vacuum they are equal to 0. The scalars Ψ_0 and Ψ_4 are related to ingoing and outgoing gravitational radiation, so in a system that emits GWs, Ψ_4 is the relevant quantity.

As an example, in Cartesian coordinates we have that the non-zero components in the Rie-

mann tensor are

$$\begin{aligned} R_{txtx} &= R_{txxz} = -R_{tyty} = -R_{tyyz} = -\frac{1}{2}\ddot{h}_+, \\ R_{txty} &= R_{txyz} = -\frac{1}{2}\ddot{h}_\times. \end{aligned} \quad (1.1.22)$$

A convenient choice of basis vectors (that fulfill the relations (1.1.19)) in NR would be

$$\begin{aligned} l^\mu &= \frac{1}{\sqrt{2}}(1, 0, 0, 1) \quad ; \quad n^\mu = \frac{1}{\sqrt{2}}(1, 0, 0, -1); \\ m^\mu &= \frac{1}{\sqrt{2}}(0, -i, 1, 0) \quad ; \quad \bar{m}^\mu = \frac{1}{\sqrt{2}}(0, i, 1, 0); \end{aligned} \quad (1.1.23)$$

which when plugged in Eq. (1.1.21) we get

$$\Psi_4 = \ddot{h}_+ - i\ddot{h}_\times. \quad (1.1.24)$$

This result can be further extended for spherical coordinates as the transversal polarizations of GWs propagating in the outgoing radial direction, where the same expression (Eq. (1.1.24)) holds. The relation between the strain and the Weyl scalar can thus be computed from this equation with the following double integral

$$h_+^{lm}(r, t) - ih_\times^{lm}(r, t) = \int_{-\infty}^t du \int_{-\infty}^u dv \Psi_4^{lm}(r, v), \quad (1.1.25)$$

where we can see that the Weyl scalar can be expressed in terms of its modes in the same way as done in Eq. (1.1.17).

1.2. Black hole dynamics

In this project, we are required to work with non-static BHs that create a dynamical spacetime around them. Therefore, it is first needed to understand some key properties to fully grasp the results in the following chapters.

1.2.1. Kerr black holes

In a strong-gravity regime, a large mass (for example a BH) moving through spacetime — either rotationally or linearly — “pulls” spacetime along its motion. This phenomenon is called frame-dragging. The rotational frame-dragging can be understood analytically with the Kerr metric, while the linear frame-dragging will be described later (more concretely in Sec. 2.5.). When we consider a spinning BH fixed at some point in space, one can find the analytical solution to

Einstein's equations that describe the metric around these so-called Kerr black holes:

$$ds^2 = - \left(1 - \frac{2Mr}{\Sigma}\right) dt^2 - \frac{4aMr \sin^2 \theta}{\Sigma} dt d\phi + \frac{\Sigma}{\Delta} dr^2 + \Sigma d\theta^2 + \left(r^2 + a^2 + \frac{2Ma^2r \sin^2 \theta}{\Sigma}\right) \sin^2 \theta d\phi^2, \quad (1.2.1)$$

where $\Sigma = r^2 + a^2 \cos^2 \theta$, $\Delta = r^2 - 2Mr + a^2$, M is the mass of the BH and $J/M = a$ is the specific angular momentum per unit mass. When $a = 0$ we recover Schwarzschild's solution for static black holes

$$ds^2 = - \left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2). \quad (1.2.2)$$

For static black holes, we can see there is a singularity at $r_s = 2M$, which is called the event horizon or Schwarzschild radius, which is the surface that determines the inner region from where light cannot escape. If we do a more in-depth analysis in the case of Kerr black holes, we can see that the singularity appears for $\Delta = 0$ which describes two horizons

$$r_{\pm} = M \pm \sqrt{M^2 - a^2}. \quad (1.2.3)$$

The most relevant result we can get from this expression in relation to our simulations is that, to have an event horizon, we need a non-negative argument inside the square root in Eq. (1.2.3), which restricts the angular momentum per unit mass to have a maximum value of $a_M = M$. This implies that $J/M = a = \chi M$, with $0 \leq \chi \leq 1$, and for all the results in this project related to spin, we will use the parameter χ as the one that defines the spin of a BH.

Moreover, Kerr black holes have an additional surface around them called ergosphere given by

$$r_{e,\pm} = M \pm \sqrt{M^2 - a^2 \cos^2 \theta}. \quad (1.2.4)$$

Thus, any particle located at radii $r_+ < r < r_{e,+}$ is forced to co-rotate with the BH spin.

1.2.2. GR orbits

In the study of black hole (BH) dynamics, two primary configurations arise based on the system's initial energy. If the total energy is positive ($E > 0$), the BHs are initially unbound and follow hyperbolic orbits, resulting in a (close) hyperbolic encounter (CHE). Conversely, if the total energy is negative ($E < 0$), the system is bound, and the BHs orbit each other with an eccentricity in the range $0 \leq e < 1$, describing (quasi-)elliptical trajectories. These general configurations are illustrated conceptually in Fig. 1, which shows the effective potential in general relativity. While this potential offers a simplified and qualitative picture, the actual dynamics explored in our simulations are considerably more intricate, incorporating spin effects, relativistic corrections, and GW emission.

A key difference between GR and Newtonian gravity lies in the non-linear nature of Einstein's equations, which allows for the emission of energy in the form of GWs when massive objects, like BHs, undergo acceleration — particularly during close passages where trajectories and velocities

change rapidly. This GW emission acts as a dissipative mechanism, extracting energy and angular momentum from the system. The process is somewhat analogous to Bremsstrahlung radiation in electromagnetism, where accelerated charges emit photons. As a result of this energy loss, the

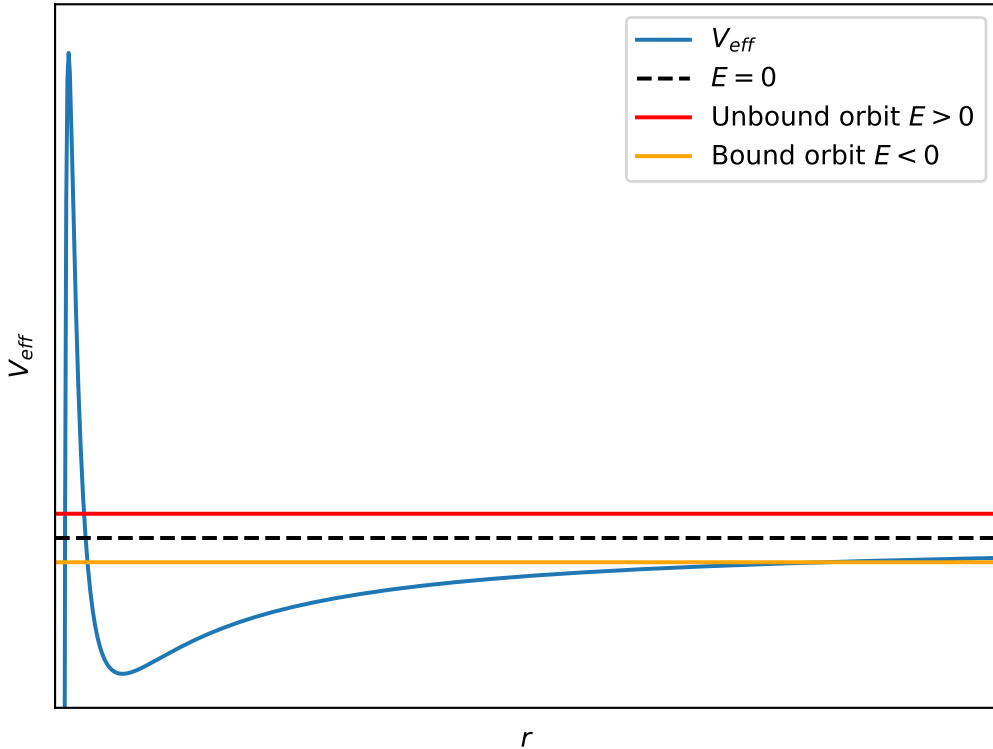


FIGURE 1: The two main configurations discussed in this project. The effective potential is the simplest one that appears in General Relativity, but essentially it is similar to the Newtonian case when viewed at large distances.

system’s effective energy decreases, causing the BHs to “fall” deeper into the potential well. Over time, this leads to a progressive reduction in orbital eccentricity as the BHs spiral closer together, forming a new BH from the fusion of two, radiating away GWs as well. In some extreme cases, particularly during close hyperbolic encounters, the energy loss is significant enough to transform an initially unbound orbit ($E > 0$) into a bound one ($E < 0$). This phenomenon, known as a dynamical capture (DC), can ultimately lead to a merger, even though the system started in an unbound configuration. Dynamical captures are particularly relevant in dense astrophysical environments such as globular clusters, galactic nuclei, or stellar clusters, where BHs interact frequently at high relative velocities.

Understanding these processes is crucial for predicting the formation of BBH mergers, especially those involving highly eccentric orbits or unexpected capture scenarios. Such events could produce distinctive GW signatures, potentially observable by detectors such as LIGO, Virgo, KAGRA, or future observatories such as LISA. These observations would provide invaluable insights into the astrophysical environments in which such encounters occur and the role of spin, eccentricity, and relativistic effects in shaping BH interactions.

1.3. Numerical Relativity formalism

As we have already mentioned, the dynamics of binary systems are very non-linear, which is why solving Einstein's equations with conventional methods would be a complex (if not impossible) task. Therefore, we resort to Numerical Relativity to solve binary coalescences and hyperbolic encounters.

However, NR simulations are highly computationally expensive to run, and thus waveform models are common to analyze LVK data, as they take seconds to run. However, these models are still not sufficiently accurate, and their performance will worsen when detector sensitivity improves. This is why Numerical Relativity waveforms are still indispensable to test Einstein's theory, also noting that close hyperbolic encounters (CHEs) and dynamical captures (DCs) — the main topics of this project — yet have only a few waveform models [10, 16, 33, 34, 58], in comparison with CBC² waveform models, and have not been detected by LVK yet (although some events are promising candidates [35]).

1.3.1. The 3+1 ADM and BSSN formalisms

The basis upon which Numerical Relativity [62] is constructed is the ADM (Arnowitt-Deser-Misner) formalism [13]. It consists in slicing the 4-dimensional spacetime in 3+1 components, as the covariant form of Einstein's equations is not suitable for time-evolving a general-relativistic system. This procedure is called foliation.

With the foliation method, spacetime can be split into a family of spacelike hypersurfaces Σ , each labeled by a time parameter. A congruence of timelike curves³, representing our observers (or chosen coordinate system), intersects these hypersurfaces. This congruence of curves is described by the vector field $t^a = \alpha n^a + \beta^a$, where n^a is the unit timelike vector, normal to the said hypersurfaces, α is the lapse function quantifying the proper time experienced by Eulerian (orthogonal) observers, and β^a is the shift vector that accounts for the spatial displacement of the observers' worldlines relative to n^a as they move from one hypersurface to the next.

In this framework, any four-dimensional object can be decomposed into its 3+1 components. The coordinate choice naturally defines a spatial projection tensor $\gamma^a_b \equiv \delta^a_b + n^a n_b$, which projects quantities onto the hypersurfaces Σ . Using this, any four-dimensional tensor can be split into spatial parts — by applying the projector — and temporal parts — by contracting with n^a .

Once all of the referenced parameters have been defined, and our coordinates selected, we can define the line element of this spacetime as

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu = -\alpha^2 dt^2 + \gamma_{ij} (dx^i + \beta^i dt)(dx^j + \beta^j dt). \quad (1.3.1)$$

Within this 3+1 framework, the *induced metric* γ_{ij} on the spatial hypersurfaces is the primary dynamical variable, as both the lapse function α and the shift vector β^i are determined by our choice of coordinates (we have gauge freedom). The system of equations in the ADM formulation

²CBC is the acronym for Compact Binary Coalescence

³A congruence (of curves) is a set or family of curves in which a given curve passes through a certain point in spacetime $\mathcal{O}$ solely by this curve and non other in the family. Thus, the curves do not intersect [64].

changes accordingly to our choice of coordinates, preserving 3+1 covariance under spatial or time transformations.

To further describe the geometry of the hypersurfaces and their embedding in the full space-time, differential geometry introduces an additional tensor, called the *extrinsic curvature* K_{ij} . This tensor carries significant geometric meaning, as it quantifies how the induced metric evolves along the flow of the normal observers.

The extrinsic curvature is defined through the *Lie derivative* $\mathcal{L}_n$, which generalizes the directional derivative along the normal vector n^a within a manifold. The specific expression for the extrinsic curvature is given by

$$K_{ij} \equiv -\frac{1}{2\alpha}\mathcal{L}_n\gamma_{ij} = -\frac{1}{2\alpha}(\partial_t - \mathcal{L}_\beta)\gamma_{ij}, \quad (1.3.2)$$

where one can see that $\mathcal{L}_n = \partial_t - \mathcal{L}_\beta$, with $\mathcal{L}_\beta$ denoting the Lie derivative along the shift vector β^i . More specifically, K_{ij} measures how the spatial geometry changes as one moves in time along the congruence of normal observers.

Once all of these tensors and parameters have been introduced, we obtain the formalism (ADM) which results in a system of partial differential equations (PDEs) with both hyperbolic and elliptic characteristics. In this formulation, Einstein's equations can be written as:

$$\partial_t\gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta\gamma_{ij}, \quad (1.3.3)$$

$$\partial_t K_{ij} = -D_i D_j \alpha + \alpha \left(R_{ij} + K K_{ij} - 2K_{ik} K^k_j \right) + \mathcal{L}_\beta K_{ij}, \quad (1.3.4)$$

where D_i is the covariant derivative associated with γ_{ij} ; R_{ij} is the three-dimensional Ricci tensor of the spatial slice; and K is the trace of the extrinsic curvature, $K = \gamma^{ij} K_{ij}$.

Although the ADM formalism was widely used in the early stages of Numerical Relativity due to its simplicity, it was later found to be unsuitable for computational applications in three-dimensional scenarios due to the emergence of several numerical instabilities. By the 1990s, it became clear that the ADM system, under standard gauge choices, lacks the mathematical robustness required for reliable evolution, leading to persistent numerical instabilities. These issues stem largely from the so-called momentum and Hamiltonian gauge conditions, since any numerical errors that deviate even slightly from these constraints accumulate over time, leading to instability, as we obtain unphysical, growing modes for our GW wave equation.

To address this issue, a conformal metric $\tilde{\gamma}_{ij}$ with unit determinant and a trace-free conformal extrinsic curvature $\tilde{A}_{ij}$ are introduced as

$$\tilde{\gamma}_{ij} = \chi \gamma_{ij} \quad ; \quad \tilde{A}_{ij} = \chi \left(K_{ij} - \frac{1}{3} \gamma_{ij} K \right). \quad (1.3.5)$$

However, this transformation alone is not enough to fix the numerical instabilities — in fact the convergence worsens. It only works when paired with the additional and appropriate gauge conditions for two new independent variables ($\tilde{\gamma} = 1$, $\tilde{A} = 0$) we introduce to remove the unphysical growing modes in the GW time-evolution, obtaining a significant improvement in the mathematical structure of the system, ensuring more controlled and stable numerical evolution.

In this Thesis, we will not show how these instabilities are removed by this “modified ADM” formalism — the BSSN (Baumgarte-Shapiro-Shibata-Nakamura) formalism —, but are shown in the references [15, 77].

There are alternative methods, such as the harmonic slicing [15], nevertheless, it is not worth it going deeper into this topic, as it is only necessary to know the basics of NR to understand the results and analysis of this Thesis. A very neatly explained — and in much more detail — introduction to NR was recently written by Dr. Carlos Palenzuela [62].

1.3.2. NR initial conditions

The initial conditions of a BBH simulation can be determined by the masses m_1, m_2 , the linear momenta $\vec{p}_1, \vec{p}_2$ and the spins $\vec{\chi}_1, \vec{\chi}_2$ of each black hole. An important quantity to introduce is the mass ratio q which is commonly defined as

$$q \equiv \frac{m_2}{m_1} ; m_1 \geq m_2 , 0 \leq q \leq 1. \quad (1.3.6)$$

Although the opposite convention ($q \geq 1$) is also used in the literature, we will use the first one, as it is the one used in the works referenced in this Thesis.

Moreover, we use the center of mass (CM) frame in our simulations, to make our results easier to interpret. This implies that the initial momenta must have opposite directions between both BHs $\vec{p}_1 = -\vec{p}_2$ (see Fig. 2).

In spite of the apparent simplicity of setting initial conditions in our simulations, it is needed as well to define the metric quantities in Eq. (1.3.1) from the ADM formalism, which is a non-trivial task. Therefore, one needs to use some method that generates proper initial conditions from the initial data.

A widely used method in NR codes is the so-called puncture initial data, based on the Bowen-York prescription [17], allowing for inclusion of angular momentum and spin, which previous methods in the dawn of Numerical Relativity could not handle.

On another note, the presented details of initial conditions are useful to start off a simulation, but these do not exactly correspond to the state of a binary black hole, following a natural evolution of the system. The initial conditions are based on the assumption of conformal flatness, which does not accurately represent the relaxed state of black holes possessing angular momentum. As a result, when the simulation starts, the system is initially disturbed in an unphysical way. It then undergoes a period of adjustment until it settles into a more natural, numerically stable evolution. This early phase of the simulation produces a burst of gravitational waves commonly referred to as *junk radiation* [44, 53].

Finally, all of our simulations are defined in a grid. Since computational power is limited, we need to set some boundaries, and to avoid gravitational-waves propagating towards the center we define Sommerfeld radiative conditions, which makes the boundary behave as outgoing spherical waves. Additionally, in the following subsection we will comment how to improve the efficiency of our simulations when defining the aforementioned grid.

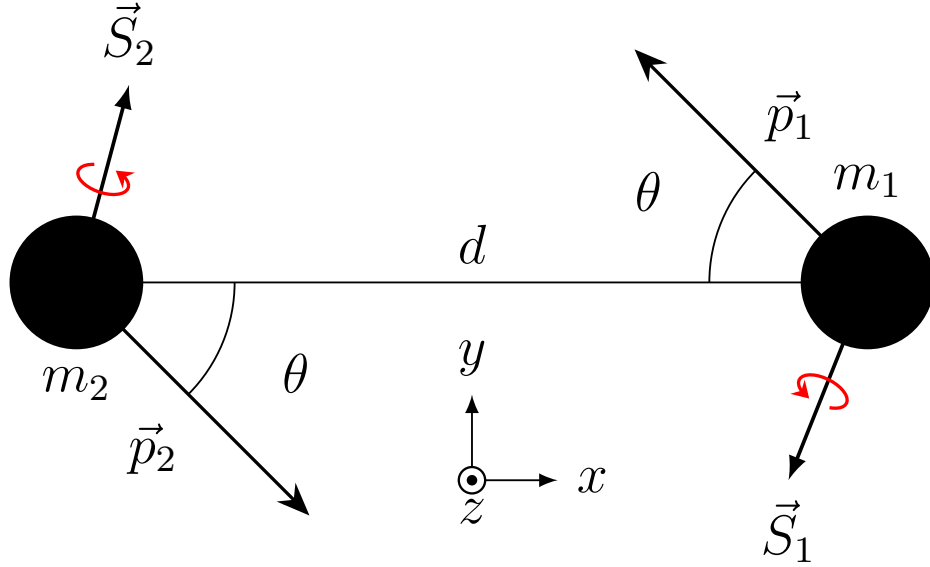


FIGURE 2: General initial configuration used throughout this work, where $m_1 = m_2$, $\vec{p}_1 = -\vec{p}_2$. In the study of CHEs we will include both parallel and perpendicular spins to $\vec{L}$, and in the analysis of DCs we will include solely non-spinning black holes.

1.3.3. Adaptive mesh refinement

BBH simulations, especially the ones dealing with hyperbolic encounters, require a simulation length of the order of hundreds of Schwarzschild radii of the black holes, in order for the boundaries to not significantly affect the physical results, which occurs when the boundary is sufficiently far away from the most dynamically changing area.

This, and the fact that near the horizons the resolution must be greater than a hundredth of a Schwarzschild radius, the computational cost of uniformly discretizing the simulation volume in a conventional way would be immense. Therefore, we need, in order to increase the efficiency of the program, that the most changing regions that most strongly experience the dynamics of the BHs have a greater degree of discretization than those that do not suffer as many alterations. Thus, we create an adaptive mesh refinement (AMR) around the BHs where the tensors and components that we must calculate have very high precision, while far away from them, where not as much precision is needed, the discretization is of a smaller degree. And so, we create a grid for the entire space and as we approach each BH we add some additional grid levels depending on what the simulation requires. This is visually shown in figure 3.

In this figure, we depict an example with 5 refinement levels. The coarsest level covers the full simulation volume, with a mild resolution in both space and time. The next level covers a portion of the full volume, with greater resolution, and successive levels will describe a portion of the previous levels with an even greater resolution. Consecutive levels are generally taken to be half the size and double the resolution than the previous one in Numerical-Relativity simulations.

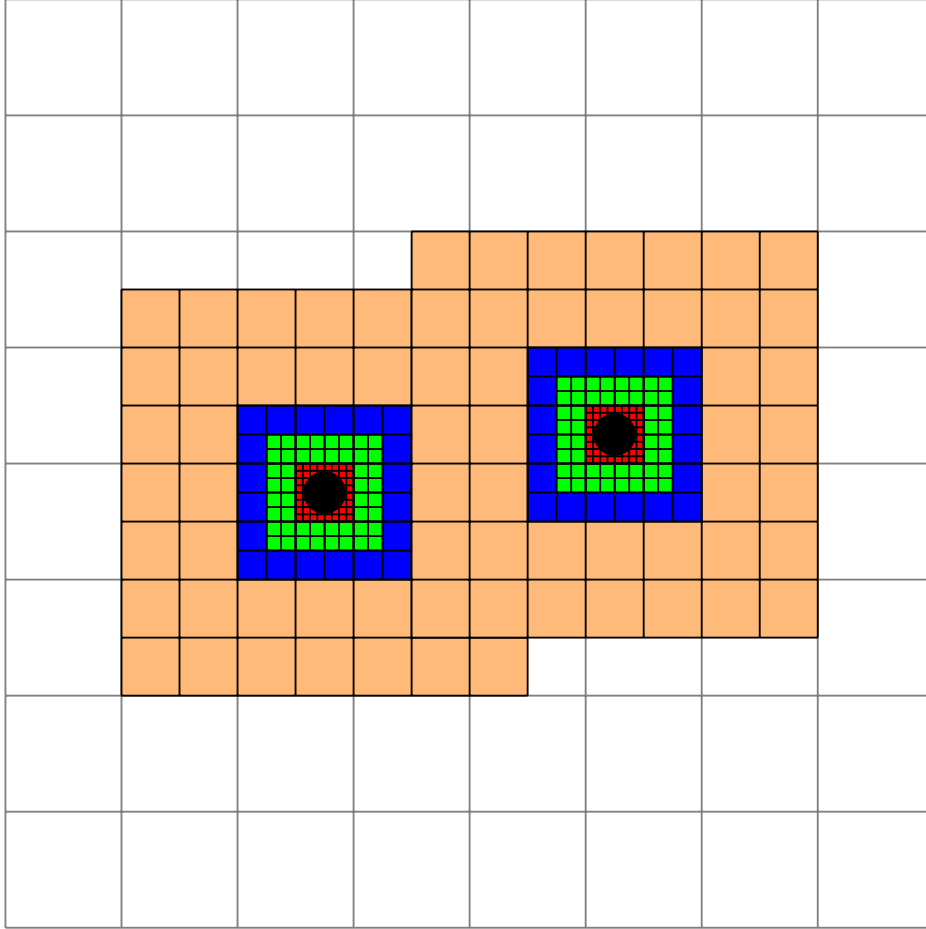


FIGURE 3: Visualization of the mesh refinement around two black holes.

1.3.4. Apparent horizons

Finding black hole horizons is key to accurately describe the evolution the full BBH system in a NR simulation. Nevertheless, when we talk about horizons in the context of a dynamical spacetime, the concept of event horizon is no longer useful since an event horizon is a global property of the total spacetime, only useful for stationary scenarios. In order to measure the properties of evolving and merging black holes one needs a definition of surface that locally separates outgoing and ingoing light rays which is called apparent horizon.

The equation that defines the apparent horizon surface is defined as

$$q^{ij}(K_{ij} - D_i s_j) = 0, \quad (1.3.7)$$

where s_i is a spacelike vector orthogonal to the surface of the apparent horizon, D_i the covariant derivative, $q_{ij} = \gamma_{ij} - s_i s_j$ and K_{ij} the extrinsic curvature. A more detailed description on how to define and construct the apparent horizon of a BH is defined in [8].

When one computes the surface of the apparent horizon, one can calculate the area of the

black hole, with which one can give a value for its (horizon) mass [19] as

$$m_H = \sqrt{\frac{A}{16\pi} + \frac{4\pi J^2}{A}}. \quad (1.3.8)$$

We can redefine some variables in Eq. (1.3.8) that will allow us to better comprehend the nature of BH masses. First, the ADM mass of a BH — the mass it has as an isolated gravitational system in asymptotically flat spacetime — would be $m \equiv m_H^4$, while the so-called irreducible mass — the portion of the BH's mass that cannot be extracted, even through processes like energy extraction via the Penrose process or black hole mergers [74] — would be $m_{\text{irr}} \equiv \sqrt{A/(16\pi)}$. Thus, the previous equation will transform into

$$m_H = \sqrt{m_{\text{irr}}^2 + \frac{J^2}{4m_{\text{irr}}^2}}, \quad (1.3.9)$$

which establishes the relation between ADM and irreducible masses.

1.3.5. Measuring spin

The spin of a BH in a dynamical context is a gauge dependent variable, hence, it has several definitions. Only when black holes are well separated, up to the point when they become isolated, all definitions of spin coincide. One of the ways to estimate BH spins, is by using the shape of the BH apparent horizons. One can easily find their spins with the following expression (as shown in [6])

$$\frac{C_p}{C_e} = \frac{1 + \sqrt{1 - (a/m)^2}}{\pi} E\left(-\frac{(a/m)^2}{(1 + \sqrt{1 - (a/m)^2})^2}\right), \quad (1.3.10)$$

where C_p and C_e are the polar and equatorial circumferences — measured with their respective geodesics — respectively. The variable $a = J/m$ (defined back in Subsec. 1.2.1.), where J is the specific angular momentum of the black hole and m its ADM mass. Moreover, $E(x)$ is the complete elliptic integral of the second kind, defined in the following way

$$E(x) = \int_0^{\pi/2} d\theta \sqrt{1 - x \sin^2 \theta}. \quad (1.3.11)$$

In our case, it will not be needed to use the previous expression, as the post-processing tools we employ to carry out the analysis of each simulation computes the values of the horizon and the elliptic integral directly to find the value of a .

These are the basics of Numerical Relativity that will help better understand this work. As we have mentioned, solving the presented equations regarding the ADM and BSSN (in Subsec. 1.3.1.) formalism from scratch is a colossal task (as well as manually computing the apparent horizons or spins from Eqs. (1.3.7) and (1.3.10)), and therefore, as we will show in full detail in the following section, we use a collection of software components and tools to effectively simulate the different general-relativistic astrophysical scenarios.

⁴Sometimes the ADM mass is written as m_{ADM} , but we avoid this notation to be as concise as possible.

2. Preliminary setup and event dynamics

In this chapter, we aim to set the explicit details of our simulations, along with including some general remarks on the behavior that the events that will be analyzed in the next chapters follow.

2.1. The Einstein Toolkit

The Einstein Toolkit (ETK, [43]) is a set of tools used to create high-level simulations in NR. The high complexity of Einstein’s equations in the context of binary black holes practically forces us to use the ETK in a cluster (*Hydra IFT*, in our case). A cluster is a group of computers that work in parallel to perform complex calculations and heavy data storage more efficiently, therefore, what a conventional computer cannot do, our cluster’s High Performance Computing (HPC) can deal with it without any problems.

The ETK has a large number of modules called *thorns* — where each thorn has a specific role — that work in the *Cactus* environment where all the coordinates, interpolators, boundary conditions, etc. are defined and which are essential for the correct operation of our simulations.

2.1.1. Important thorns

In general, most of the ETK components use the CACTUS Computational Toolkit, which acts as the environment where all the modules function properly as a coordinated ensemble. The following is a list of some of the most relevant thorns we use throughout this project. This list aims to give an intuition of how the ETK works and its tools and functionalities, but a more in-detail description of all these capabilities and roles can be found in the ETK website [43] and in the references [13, 25, 62, 83].

- The AMR grid infrastructure is provided by the CARPET thorn [42, 75], which separates the technical part of the simulation from the other thorns related to “pure physics”, for more efficient calculations. It also distributes the grid functions among the parallel processes, coordinating their communication and regridding when requested, following the black-hole motion.
- Complementing the previous thorn, the LLAMA code [65] makes it possible to have a domain consisting of overlapping patches caused by the AMR.
- The black hole positions — which are used by the AMR regridding — are tracked by the PUNCTURETRACKER thorn.
- The black hole apparent horizons are found via the AHFINDERDIRECT thorn [80], requiring an initial guess for the horizon shapes.
- The ADMBASE thorn provides the core infrastructure for implementing GR using the 3+1 ADM formalism on a 3D grid. It defines key variables — such as the 3-metric, extrinsic curvature, lapse, and shift vector — and parameters to control their evolution. These variables enable communication between components (thorns) handling initial data, evolution, and analysis, and can also bridge with alternative formalisms through appropriate transformation routines.

- Einstein’s equations are evolved using the BSSN formalism by the MCLACHLAN code [31, 69] — which interoperates through the ADMBASE and TMUNUBASE (whose job is to define the stress-energy tensor throughout the simulation) interfaces —, generating the necessary thorns by using KRANC [45, 46] as the corresponding Mathematica application which turns a tensorial description of a time-dependent partial differential equation into a module for the Cactus Computational Toolkit. KRANC is designed to take a simple continuum description of a problem and generate a highly efficient and portable code.
- The TwOPUNCTURES thorn [11] provides the initial puncture data, using optimized spectral interpolation [63]. It is not the only option for initial data generation, but it is the most widely used one in BBH simulations.
- The aforementioned Sommerfeld radiative conditions are generated by the NEWRAD thorn [7].
- The QUASILocalMEASURES thorn [32, 76, 82] computes mass and spin multipoles using the isolated and dynamical horizon formalism, along with various other proposed formulas for quasilocal mass, linear momentum, and angular momentum. Given a specified surface — such as a horizon or large coordinate sphere — the thorn can also calculate quantities like Weyl and Ricci scalars and the three-volume element of the horizon world tube, making it a key tool in numerical relativity.
- The WEYLSCAL4 thorn calculates the Weyl scalar as the outgoing gravitational-wave radiation of an asymptotically flat system via KRANC.

Finally, the SimFactory [79] provides an environment for the configuration, building, submission and management of our simulations in a user-friendly manner. It allows us to control all the capabilities of the ETK without delving into the most technical aspects of the programme.

2.1.2. Processing and post-processing

Once a simulation has finalized, we can obtain the output and analyze it. However, the multiple output directories and complicated output structure of the simulation makes the direct result processing and post-processing not very straightforward. Additionally, representing variables defined on an AMR grid as simple arrays overlooks the underlying grid structure, making certain operations difficult or even unfeasible. This creates substantial friction when trying to explore the scientific insights within a simulation. However, there is a convenient, user-friendly way to analyze our simulations: Kuibit [18], the ETK python package for processing and post-processing our simulations.

With this python package we can obtain — without worrying much about how the data is stored — the direct output of the simulation (*processing*), such as the relevant Weyl scalar and the BH positions, masses, spins, apparent horizons, etc. We can also derive the indirect output (*post-processing*) such as the strain from the Weyl scalar or the 2-dimensional data to visualize the GW propagation, to give some examples.

Kuibit is, thus, the way to extract the full information of a simulation — either directly from the output or applying some numerical operations over the output — with simple commands and in an intuitive manner, significantly simplifying the simulation analysis and being very well documented, which lowers the learning barrier to use the tool.

2.2. Explicit details of the simulations

As mentioned in the previous section 2.1., we employ the ETK with all the referenced thorns, modules and programs to run, analyze and post-process the simulations that interest us.

The initial configurations used in the simulations of the next chapters follow those of [47], with the primary difference being the inclusion of initial black hole spins (see Fig. 2) in some simulations, as we will mention later. We restrict ourselves to equal-mass binaries, $m_1 = m_2 \equiv m$, with symmetric momenta, $|\vec{p}_1| = |\vec{p}_2| \equiv p$. Working in geometrized units ($G = c = 1$), where all quantities are expressed in terms of mass, we normalize the total mass to $M \equiv m_1 + m_2 = 2m = 1$. To generalize the results for arbitrary total mass M , one can simply rescale all quantities accordingly, expressing them as t/M , p/M , x/M , etc.

In the third chapter, our main focus is on configurations where the black hole spins are aligned with the orbital angular momentum $\vec{L} \equiv L\hat{z}$. However, we also explore some cases with spins perpendicular to $\vec{L}$ to highlight additional relevant effects. Black hole spins are characterized using the dimensionless spin vector $\vec{\chi} = \vec{S}/m^2$, with magnitude $\chi = |\vec{\chi}|$, which satisfies $0 \leq \chi \leq 1$. This definition will be used consistently throughout the text.

During the fourth chapter, we will only study dynamical captures of non-spinning BHs, so the initial configuration will be identical to the one in [47]. Or, in other words, the one presented in Fig. 2 but with $\vec{S}_1 = \vec{S}_2 = \vec{0}$.

The grid setup matches that used in [47]. We employ nested refinement levels with half-lengths given by 0.75×2^n , for $n = 0, 1, \dots, 6, 8, 9, 10$. The grid spacing at each level is defined as $2^n \times \Delta x_{mr}$ for $n = 0, 1, \dots, 9$, where Δx_{mr} corresponds to the resolution of the most refined grid.

Three different resolutions are used in this work:

- *Medium resolution*: $\Delta x_{mr} = \frac{3}{200}M$.
- *High resolution*: $\Delta x_{mr} = \frac{3}{256}M$.
- *Very high resolution*: $\Delta x_{mr} = \frac{1}{104}M$.

The *medium* and *high* resolutions follow the conventions of [47], while the *very high* resolution is introduced here for cases involving large spins, where significant discrepancies were observed between the other two resolutions. These differences are most pronounced for (initial) spin magnitudes $\chi > 0.6$, as will be discussed later.

2.3. General behavior of CHEs

During Chapter 3, we will discuss some relevant effects for CHEs. In this section, we show the general behavior for these events, in order to understand their dynamics and relevant properties.

First, we plot, in the upper panel of Fig. 4, the Weyl scalar (Ψ_4 , defined back in Sec. 1.1.). For these types of events, a single scattering is produced (as seen in the lower panel of Fig. 4), which only produces one emission of GWs. This emission is produced during the strong-gravity regime experienced by both BHs during their encounter, reaching very close distances of the order of the Schwarzschild radius [47], which can also be seen in the lower panel of the aforementioned figure.

Notice that for these events, after the CHE, the BHs are scattered with a certain deflection angle — normally greater than 360° , as they do a complete “circle” before being scattered — and never meet again. This is why sometimes these scenarios are called *scattering* events or *pure* CHEs.

Notice, as well, that we have computed the Weyl scalar at a sphere with radius $r^* = 80M$, so we account for the time it takes to arrive to this detector in the x -axis of the upper panel in Fig. 4, having in mind that GWs travel at the speed of light. In general, the Weyl scalar and strain (h) are $\propto 1/r^*$ so we compute an approximately distance-independent quantity by multiplying it by the radius of the detector. We also prefer to use the Weyl scalar instead of the strain because the first is the direct output of the simulation while the latter is obtained via Kuibit [18] with numerical integration, as seen in Eq. (1.1.25).

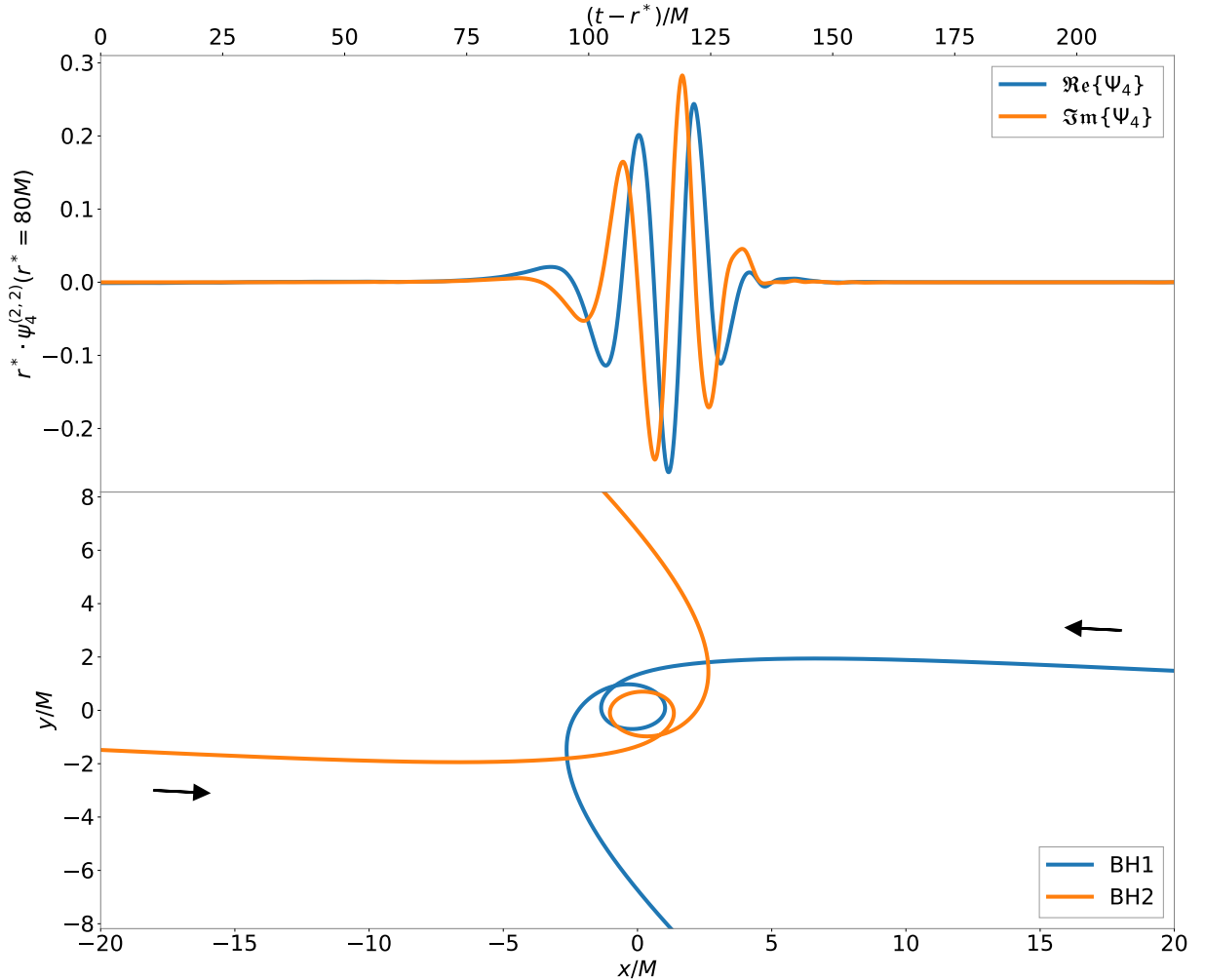


FIGURE 4: (Upper panel) Evolution of the (2,2) multipole of the Weyl scalar measured at a detector at a distance $r^* = 80M$ from the CM. (Lower panel) Trajectories of both BHs, initially hyperbolic; the arrows qualitatively indicate the $\vec{p}$ of each BH at the start of the simulation. The initial conditions are $p/M = 0.49$, $\theta = 2.864^\circ$, with zero initial spins.

In Fig. 5, in the upper panel, we see that spins ($\chi \sim 0.13$) are induced — as the BHs carry no spin initially — by the CHE. This spin-induction phenomenon is produced mainly by frame-dragging effects, but an in-depth explanation will be given in Sec. 2.5..

Additionally, in the lower panel of Fig. 5, one can observe an interesting detail: during the CHE the masses of both BHs increase slightly. This may be surprising as there is no accreted

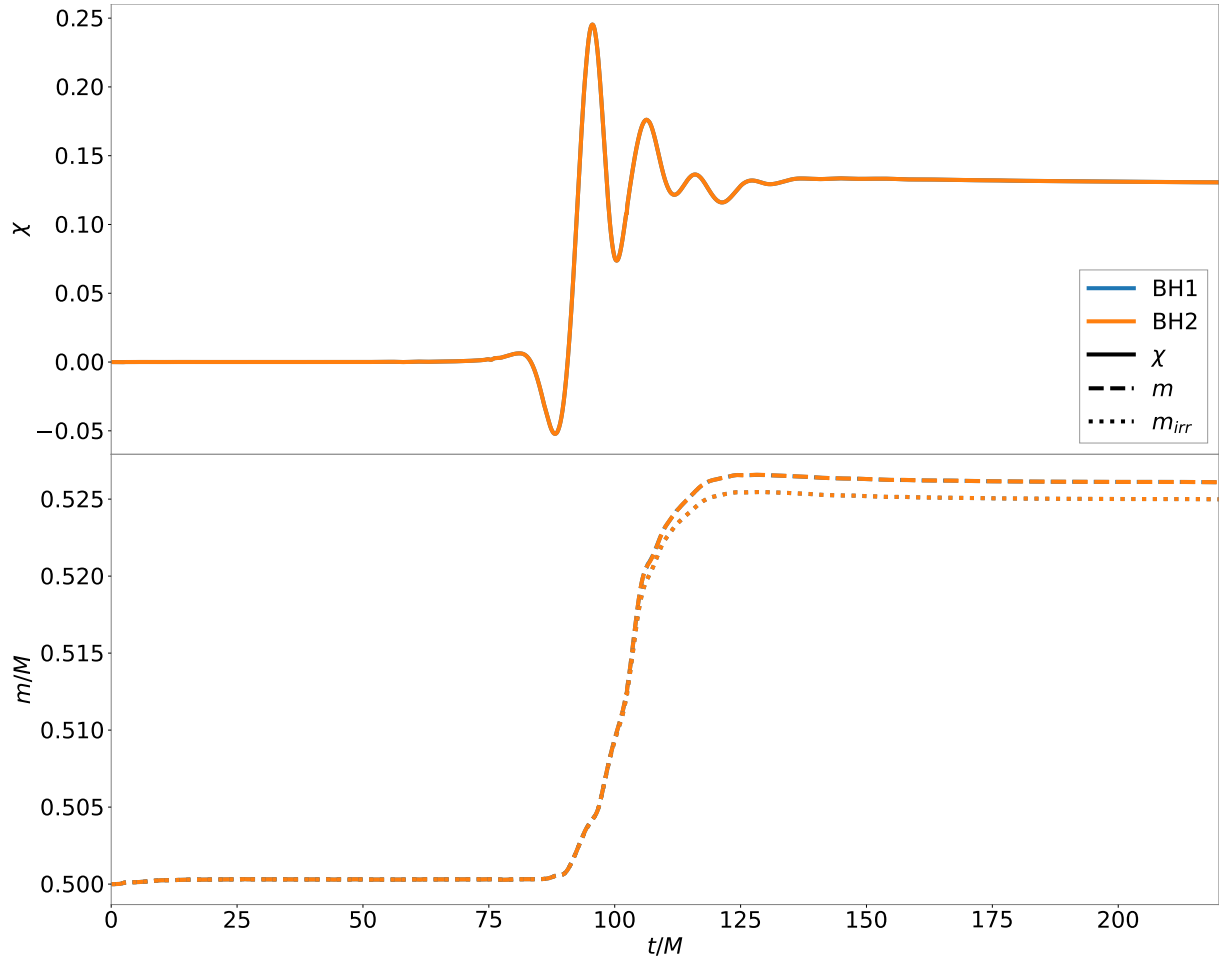


FIGURE 5: (Upper panel) Evolution of dimensionless spin parameter χ for each black hole. (Lower panel) Evolution of the mass and irreducible mass of both black holes and the merged BH. The initial conditions are $p/M = 0.49$, $\theta = 2.864^\circ$, with zero initial spins.

matter, however, previous works have shown that GWs can back-react into the horizons, causing a slight change of mass [23]. This is called tidal heating or horizon absorption. In this context, BHs act as perfect absorbers, not allowing energy to move outward from their horizons, thus in the presence of the tidal field of another compact object there occurs a change in its spin, mass and horizon area [30]. In the context of initially hyperbolic trajectories, little research has been done on horizon absorption or tidal heating, especially in the context of NR. Analytical approaches such as [23] have cast a bit of light on this topic, but here we give accurate NR-predictions for a great suite of incidence angles and momenta.

We also see that all variables in Fig. 5 evolve in the same way for both BHs, since they have equal masses and initial conditions.

In Chapter 3. we will vary the initial spins of the BHs and see how these affect the general dynamics of the BHs.

2.4. General behavior of DCs

As mentioned in the Introduction of this work, dynamical captures work in a similar fashion to CHEs, but the GW emission during the CHE binds the orbit producing a merger. In Fig. 6,

in the upper panel, we plot the Weyl scalar for a certain choice of p/M and θ along with the trajectories of both BHs (on the lower panel). Notice that two GW emissions are evident: one related with the CHE (almost identical to the one discussed in the previous section), which binds the orbit, leading to the second emission, caused by the merger of black holes. It can be observed — on the lower panel — that the first approximation (CHE) leads to a wider orbit which lasts for around $\sim 200M$, and then the BHs merge with an almost-head-on orbit.

One characteristic we will be interested in analyzing during Chapter 4. is how the initial conditions affect the amplitude of each emission. We expect that higher momenta cause a stronger emission during the CHE — as they occur faster and more aggressively, with stronger frame-dragging effects —, which is related to the order of impact parameters of the initial trajectory: the higher the p/M , the lower the incidence angles that give capture events and the stronger the emission. This will be clearly presented in the respective chapter.

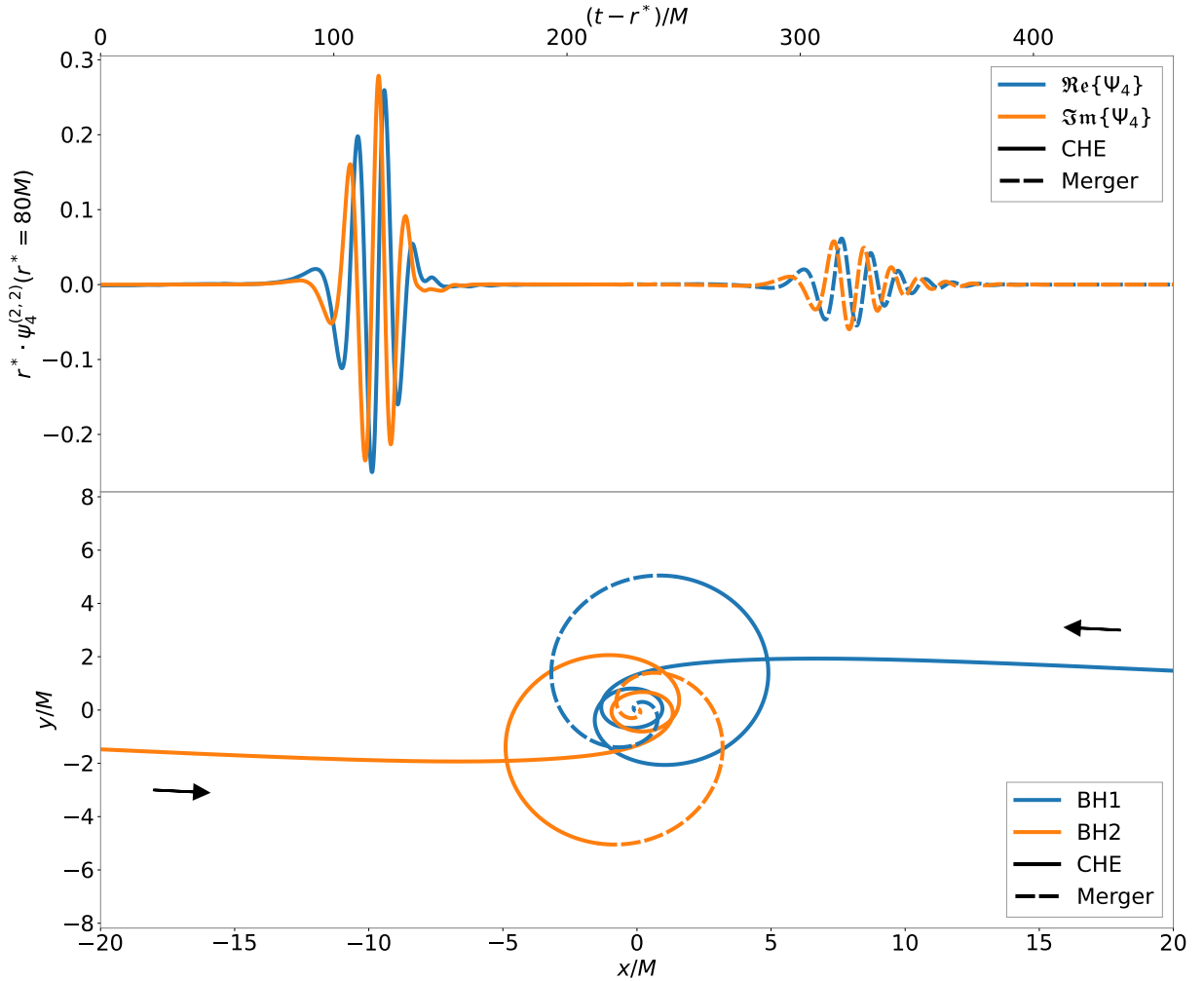


FIGURE 6: (Upper panel) Evolution of the (2,2) multipole of the Weyl scalar measured at a detector at a distance $r^* = 80M$ from the CM. (Lower panel) Trajectories of both BHs, initially hyperbolic; the arrows qualitatively indicate the $\vec{p}$ of each BH at the start of the simulation. The transition between capture and merger behavior is qualitatively shown as solid and dashed lines, respectively. The initial conditions are $p/M = 0.49$, $\theta = 2.847^\circ$.

We also plot, in Fig. 7, the evolution of spin (χ) and masses of each BH. We clearly distinguish between the final BH and the initial ones with different colors. Since we are in the CM frame

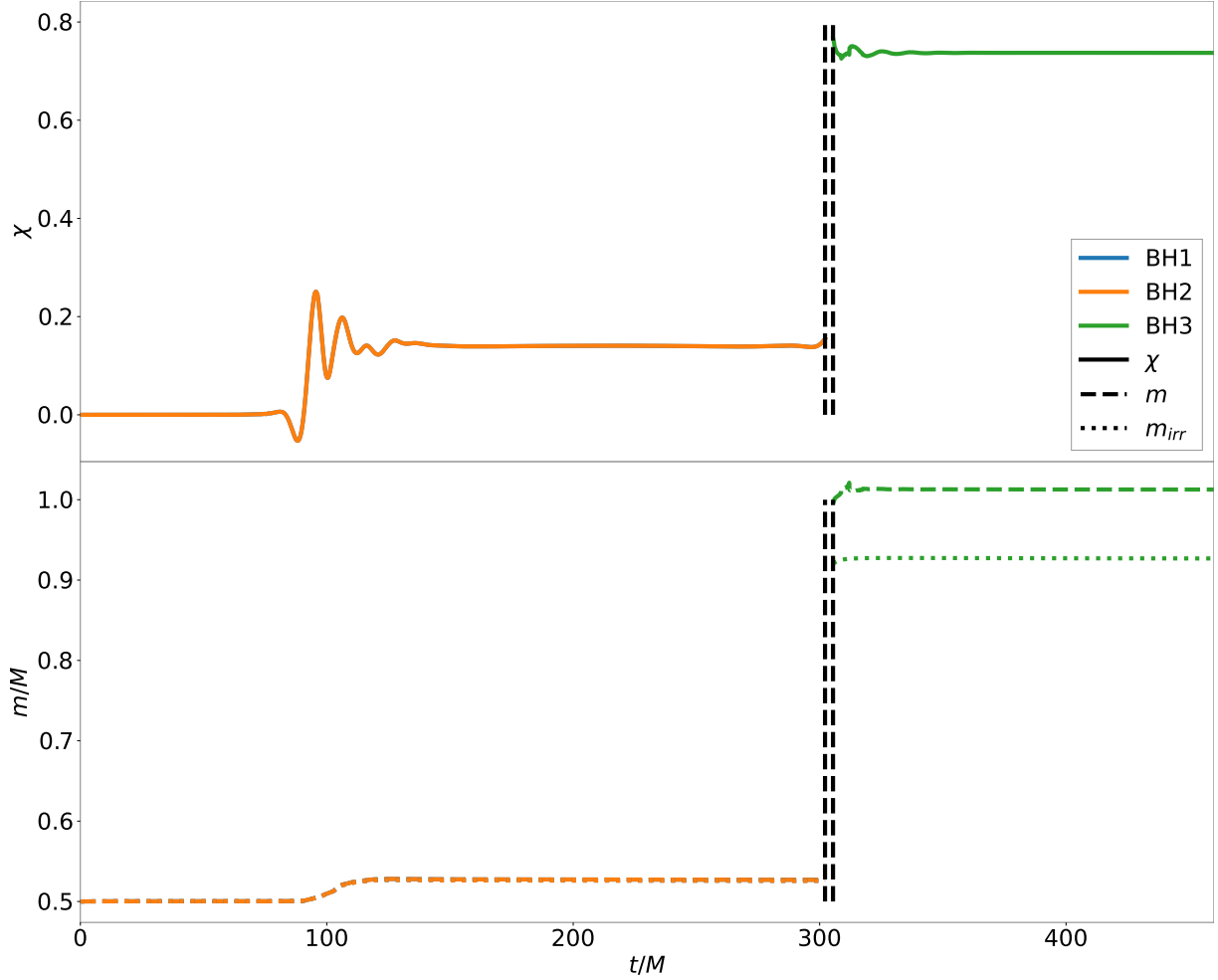


FIGURE 7: (Upper panel) Evolution of dimensionless spin parameter χ for each black hole and the final BH. (Lower panel) Evolution of the ADM mass and irreducible mass of both black holes and the merged BH. The initial conditions are $p/M = 0.49$, $\theta = 2.847^\circ$. The vertical dashed lines mark the merger of BHs.

and we have equal masses, the initial BHs evolve in the same way. Notice that during the CHE there is a noticeable spin induction, as expected from the previous section.

In the shown case, during the CHE, the increase in spin is of about ~ 0.14 while the increase in mass is $\sim 5\%$. An interesting fact is that the mass of the final BH is slightly greater than the sum of the mass of both BHs before the CHE $1.01M$, which shows that greater masses than the sum of the initial ones can be achieved in a DC even having a mass loss during the merger GW burst.

As has been seen throughout the last two sections, the behavior of pure CHEs and DCs are similar, with DCs ending up merging and producing a second GW emission. In Chapter 4, we will estimate the threshold value for the incidence angle that separates the pure CHE phenomenology from the one of DC.

2.5. Black hole interaction

In order to understand spin induction mechanisms, one must first introduce some expressions that will help us understand how two spinning black holes interact. The dynamics of these are characterized by the following Hamiltonian:

$$H \simeq H_{\text{orb}} + H_{\text{SO}} + H_{S_1 S_2}, \quad (2.5.1)$$

where the spin-orbit and spin-spin interaction terms are given by:

$$H_{\text{SO}} \simeq \frac{2G}{c^2 r^3} \vec{L} \cdot \vec{S}_{\text{eff}}, \quad (2.5.2)$$

$$H_{S_1 S_2} \simeq -\frac{G}{c^2 r^3} \left(\vec{S}_1 \cdot \vec{S}_2 - 3(\vec{S}_1 \cdot \hat{r}_1)(\vec{S}_2 \cdot \hat{r}_2) \right). \quad (2.5.3)$$

Here, $\vec{L}$ represents the system's orbital angular momentum, while $\vec{S}_j$ and $\hat{r}_j$ denote the spin and the unit position vector of each BH ($j = 1, 2$), respectively. Additionally, the effective spin is defined as:

$$\vec{S}_{\text{eff}} = \left(1 + \frac{3m_2}{4m_1} \right) \vec{S}_1 + \left(1 + \frac{3m_1}{4m_2} \right) \vec{S}_2. \quad (2.5.4)$$

Within this framework, a negative contribution from any term in the Hamiltonian implies an “attractive” effective force⁵, while a positive contribution leads to a “repulsive” interaction. Specifically, for the spin-orbit term, this *attraction* occurs when the effective spin is anti-aligned with the orbital angular momentum, whereas *repulsion* happens when they are aligned. In contrast, for the spin-spin term, aligned spins yield *attraction*, while anti-alignment leads to *repulsion*. This behavior is explored in [71] for the spin-orbit interaction, while here, we investigate whether a similar pattern emerges for spin induction in initially spinning BHs.

The system's behavior can be further understood through the concept of precession, which reflects the rotation induced in the local inertial frame. During a CHE, each BH experiences a precession with velocity $\vec{\Omega}$ caused by the strong (linear) frame-dragging the other BH produces, which translates into spin-up due to angular momentum conservation, as described for initially non-spinning cases in [47]. However, when the BHs possess initial spin, the dynamics are notably altered. A BH with negative spin generates frame-dragging that opposes the other's precession — generating a closer hyperbolic encounter —, while a positively spinning BH enhances the precession of its companion.

As a result, when both BHs have spins aligned with $\vec{L}$, their interaction produces a further hyperbolic encounter, leading to a reduced spin-up compared to the non-spinning case. Conversely, if the spins are anti-aligned, the induced spin-up is increased.

For systems with unequal initial spins, the less-spinning BH experiences a stronger spin-up due to the enhanced precession induced by the more-spinning companion. The opposite effect is observed for the BH with greater initial spin. This dynamic is illustrated in Fig. 8. Additionally, the spin-spin interaction, also depicted in Fig. 8, acts as well as a second-order effect —

⁵In general relativity, “attractive” or “repulsive” forces refer to converging or diverging geodesics, respectively.

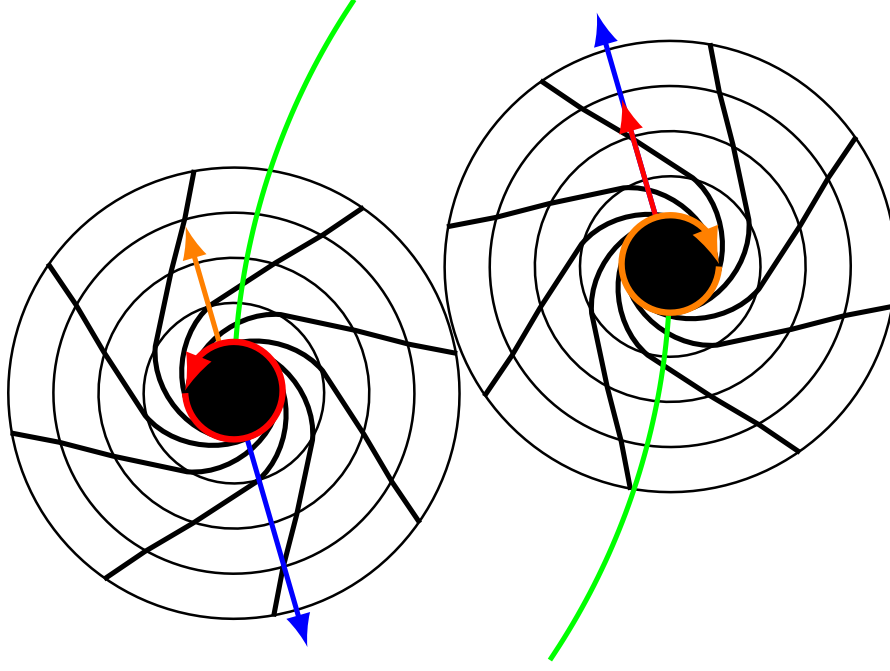


FIGURE 8: Schematic of spin induction mechanisms in close hyperbolic encounters (CHEs) with spins aligned with $\vec{L}$. The blue arrow indicates the momentum direction during the CHE, while straight arrows represent the drag effects caused by the companion BH's spin. Spin directions are shown as circular arrows, and frame-dragging effects are visualized with the BH trajectories in green.

resembling a “centrifugal” force — that becomes more pronounced as the spin difference increases. This effect further boosts $\vec{\Omega}$, contributing to a greater spin-up in both BHs.

So, we can essentially summarize the spin-induction mechanisms in the following way:

- The spin-orbit H_{SO} term in the Hamiltonian creates a *closer* or *further* hyperbolic encounter depending if the effective spin $\vec{S}_{\text{eff}}$ is anti-aligned or aligned with $\vec{L}$, producing a greater or smaller spin-up on both BHs with respect to a spinless scattering, respectively.
- Unequally spinning BHs have asymmetric induced spins because the frame-dragging of the faster-spinning BH contributes positively to the precession speed $\vec{\Omega}$ of the slower-spinning BH, and viceversa, causing a larger spin-up on the slower-spinning black hole and a lower one on the faster-spinning BH.
- For unequally spinning BHs, the spin-spin $H_{S_1 S_2}$ term in the Hamiltonian causes a boost in $\vec{\Omega}$ on both BHs which tends to a positive-second-order contribution on the spin-induction of the system that is more pronounced the greater the spin-asymmetry is.

This interplay between spin-orbit and spin-spin interactions suggests that spin induction could potentially be described analytically, for instance, using the parametrized post-Newtonian (PPN) formalism. However, since this work (more specifically Chapter 3.) focuses on numerically exploring spin induction with initial spins, we hold over the derivation of analytical expressions for future studies.

Additionally, this section forms part of a paper written by the supervisors of this Thesis and myself [72]. In the next chapter, we review the most relevant results of this publication.

3. Spin induction effects from CHEs

In this study, we concentrate on a specific dynamical feature of hyperbolic encounters: spin induction. Preceding investigations that used numerical relativity (NR) [47, 59] have demonstrated that close hyperbolic encounters (CHEs) — interactions with a small impact parameter — can induce substantial spin to initially non-spinning black holes. More recently, efforts have been made to develop analytical models that accurately describe this spin-inducing effect [23]. A precise understanding of spin induction is essential for modeling the evolution of black hole spin distributions in cluster environments and must be incorporated into future N-body simulations [61].

To date, most studies of spin induction have assumed that the black holes begin without spin, either with equal masses ($q = 1$) [59] or allowing for unequal masses ($q \neq 1$) [47]. This assumption is particularly justified in scenarios involving PBHs, which are generally expected to have negligible spin at formation. However, in dense clusters where spin-inducing encounters are likely to be more common, it is expected that black holes will acquire increasing spin magnitudes over time, after consecutive interactions. Moreover, recent work [71] has shown that initial spins can significantly influence scattering angles, which are closely tied to the resulting spin magnitudes. Therefore, it becomes necessary to explore how initial non-zero spins affect the spin induction process.

In this chapter, we investigate how introducing initial spin — both parallel and orthogonal to the orbital angular momentum — impacts the final spin states of black holes after a CHE. To do this, we have performed a series of NR simulations and we will analyze the relationship between the initial and final spin configurations. Orthogonal spins lead to spin precession⁶, a term traditionally used for CBCs, but which we adopt here to describe the change in orientation experienced by the black hole spin axes during close hyperbolic motion. Our results are then compared with simple analytical models, allowing us to interpret the spin evolution observed in the simulations.

Due to the complexity of this parameter space, our study focuses exclusively on equal-mass binary systems. This restriction allows for a more detailed examination of the role of initial spin, while the combined influence of unequal masses and spin is deferred to future investigations. Similarly, we do not explore the dependence of spin induction on the incidence angle, a parameter examined in earlier works [47, 59].

This chapter covers the main results from our paper [72], which explains in full detail the phenomenology that will be explored in the following.

3.1. Numerical results

We have conducted a series of simulations involving equal-mass black holes with initial momentum $p/M = 0.49$ and a range of incidence angles $\theta \in [2.8^\circ, 3.03^\circ]$. While our primary goal for this chapter is to study purely hyperbolic encounters, energy loss during these interactions

⁶Although spin precession was originally introduced for binary systems having shorter orbital periods than the spin precession timescale [12, 41], we have found this terminology convenient to describe the spin evolution in CHEs, since the BH spins suffer a change in the orientation of the rotation axis during this motion.

occasionally causes the black holes to become gravitationally bound, leading to a merger (DCs, see Sec. 2.4.).

Although throughout this chapter we aim to avoid DCs, they sometimes arise due to the proximity of our chosen incidence angles to the threshold between *close hyperbolic encounters* (CHEs) and DCs — particularly near the regime that induces maximal final spins. Additionally, the attractive nature of the initial spin configurations (see Sec. 2.5.) increases the chances of obtaining DCs. In such cases, while final spins are not precisely defined due to the imminent merger, we still extract spin measurements during the intermediate phase — between the initial interaction and the merger — and include them in our analysis, clearly marking them as *non-scattering* events.

Throughout this work, we use both notations χ_i and $\chi(0)$ to refer to the spin values at the start of the simulation.

3.1.1. Error analysis

In simulations involving *equally spinning black holes* (Subsec. 3.1.2.), we find that the differences between *medium* and *high* resolution (see Subsec. 2.2.) are within $\lesssim 2\%$ for maximally spin-inducing cases, and $\lesssim 4\%$ for lower spin-inducing ones. Based on this, we choose to use *medium resolution* for these simulations.

However, in the case of *unequally spinning black holes* (see Sec. 3.2.), higher resolutions are sometimes necessary. For instance, when $\chi_{\text{eff},i} \neq 0$, discrepancies between *medium* and *high* resolutions increase — likely due to additional recoil effects [71]. In some instances, a scenario appears as a DC in *medium* resolution, but resolves as a CHE at *high* resolution. As a result, we use at least *high* resolution for all cases with $\chi_{\text{eff},i} \neq 0$, while retaining *medium resolution* for the remaining configurations.

Additionally, for unequally spinning systems, we restrict our analysis to spins $\chi \leq 0.7$, as higher spins often yield differences in final spin values exceeding 5%.

All plotted data points reflect the *highest* resolution used in each simulation. We estimate error bars as follows:

- For simulations at *high* resolution, we compute the exact difference with respect to the next lower resolution.
- For simulations at *medium* resolution, we calculate a subset of representative cases to their high-resolution counterparts, then use regression techniques to estimate error bars for the remaining data points.

This approach allows us to maintain the reliability of our results without incurring unnecessary computational costs.

In general, resolution-based differences stem from the misestimation of effective “repulsive” or “attractive” forces. Lower resolutions tend to *overestimate* these effects, significantly affecting final spin values. For instance:

- When “attractive” forces dominate, low resolution typically *overestimates* final spins.
- Conversely, in scenarios where “repulsive” forces become more prominent, spins tend to be *underestimated*.

3.1.2. Equally spinning black holes

To start our analysis, we consider simulations with initial conditions $p/M = 0.49$, $\theta = 3.03^\circ$, and a range of equal spins⁷ given by $\chi_1^z(0) = \chi_2^z(0) = \chi^z \in [-0.3, 0.3]$ (recall that the structure of the initial conditions is shown in Fig. 2). All data points in this section were obtained using *medium* resolution. We exclude spins lower than $\chi^z = -0.3$, as they result in dynamical captures for this choice of incidence angle θ and momentum p/M .

Considering the references [47, 71], we expect a noticeable spin induction. Moreover, due to the presence of additional “repulsive” forces for positive spins and “attractive” forces for negative spins (see Eqs. 2.5.2 and 2.5.3), the spin induction is anticipated to be less prominent for larger initial effective spins, as stated in Sec. 2.5..

This behavior is confirmed in Fig. 9, where negative spins experience greater spin-up than positive ones. Notably, for these initial conditions, the case with the lowest spin $\chi^z = -0.3$ undergoes a spin-up of approximately ~ 0.2 , a value that, in non-spinning scenarios, would require $p/M > 0.7$ [47]. This indicates that anti-aligned spins relative to $\vec{L}$ experience stronger spin-up than aligned ones, due to the additional contributions to the Hamiltonian, particularly from Eq. (2.5.2), which also lead to a closer hyperbolic encounter.

In general, when spin is included in the initial conditions, we observe slight initial fluctuations in the spin evolution. These are due to a brief relaxation period required for the stabilization of the initial apparent horizons (see Subsec. 1.3.2.). The effect is more pronounced for larger spins, but by around $t/M = 50$, the system stabilizes and spin measurements become reliable (see Fig. 9).

In Fig. 10, we plot the final effective spin-up versus the initial effective spin. As expected from the Hamiltonian, lower values of χ^z lead to greater spin-up than higher ones. Additionally, we fit our data to a decreasing exponential function, which accurately captures the observed trend in effective spin-up:

$$\chi^z(t_f) - \chi^z = \exp \left\{ - \left[A(\chi^z)^2 + B\chi^z + C \right] \right\}, \quad (3.1.1)$$

with best-fit parameters $A = 0.61(17)$, $B = 3.50(5)$, and $C = 2.558(6)$, where the values in parentheses represent 1σ uncertainties on the last digit(s).

3.2. Unequally spinning black holes

We now explore configurations where the black holes have different initial spins. To better capture the spin dynamics, it is helpful to introduce two combined spin variables. The first is the *effective spin*,

$$\chi_{\text{eff}} = \frac{\chi_1^z + \chi_2^z}{2}, \quad (3.2.1)$$

and the second is the *weighted spin difference*,

$$\delta\chi = \frac{\chi_1^z - \chi_2^z}{2}. \quad (3.2.2)$$

⁷Since the two BHs have the same initial spins and masses, the spin induction will be the same for both [47], as shown as well in Sec. 2.3..

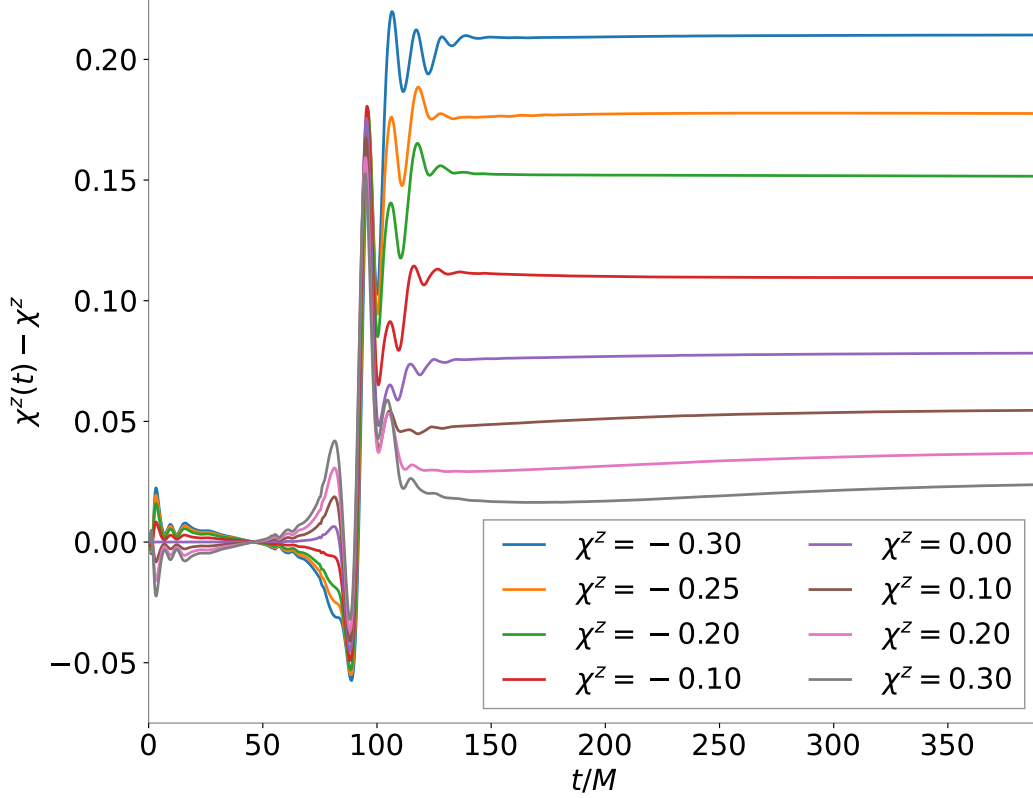


FIGURE 9: Time evolution of the spin-up for 8 cases in the interval $\chi^z \in [-0.3, 0.3]$, with $p/M = 0.49$ and $\theta = 3.03^\circ$.

These simplified forms are equivalent to more general expressions commonly used in the study of binary black holes [29, 41, 52, 67], assuming aligned spins with respect to the orbital angular momentum $\vec{L}$:

$$\chi_{\text{eff}} = \frac{\chi_1 \cos \alpha_1 + q \chi_2 \cos \alpha_2}{1 + q}, \quad (3.2.3)$$

$$\delta\chi = \frac{\chi_1 \cos \alpha_1 - q \chi_2 \cos \alpha_2}{1 + q}, \quad (3.2.4)$$

where $q = m_2/m_1$, and $\cos \alpha_{1,2}$ is the cosine of the angle between the spin vector $\vec{S}_{1,2}$ and $\vec{L}$. In our case, since we focus on equal-mass binaries ($q = 1$) and spins either parallel or perpendicular to $\vec{L}$, these general definitions simplify to Eqs. (3.2.1) and (3.2.2). Although these variables were originally developed for analyzing inspiraling systems, they also help to clarify trends in our hyperbolic encounters, especially compared to tracking each individual spin component (see Figs. 11, 12).

We examine four distinct spin configurations. In three of them, the spins are aligned with $\vec{L}$ ($\vec{S} \parallel \vec{L}$), corresponding to initial effective spin values $\chi_{\text{eff},i} = 0, -0.1$, and 0.1 . The fourth setup involves spins perpendicular to the orbital angular momentum ($\vec{S} \perp \vec{L}$). A summary of the scenarios is presented in Table 1, including the key setup parameters such as the incidence angle, spin ranges, number of simulations, and the minimum resolution used for each case. Further justification for the chosen resolution is given in Subsec. 3.1.1..

To select appropriate incidence angles θ , we begin with $\theta = 2.86^\circ$, which yields the strongest

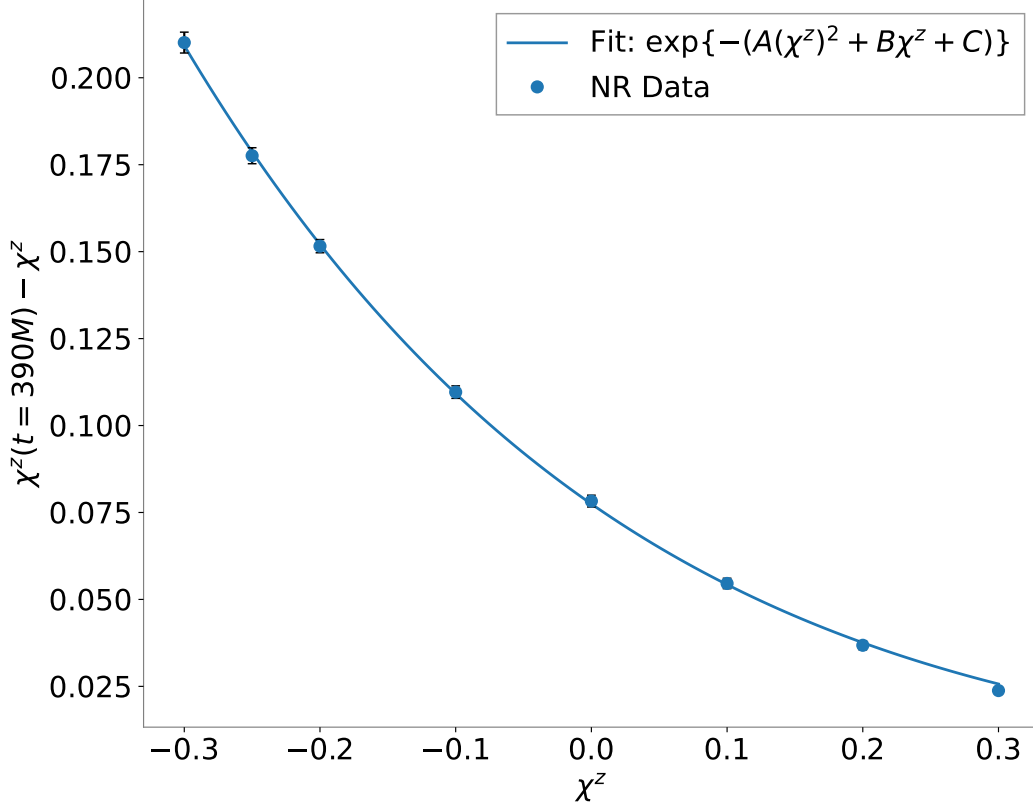


FIGURE 10: Spin-up at $t_f/M = 390$ versus the initial spin for the cases shown in Fig. 9, including an exponential fit.

TABLE 1: Overview of the spin configurations considered in Sec. 3.2., with fixed momentum $p/M = 0.49$. Each row lists the incidence angle θ , number of simulations N_S , spin initialization range $\chi_{1,2}^k(0)$, and minimum numerical resolution. Here, the superscript k indicates the spin axis: $k = z$ for aligned-spin scenarios a)–c), and $k = x$ for the perpendicular-spin scenario d).

Scenario	Label	θ	N_S	$\chi_{1,2}^k(0)$	Min. Res.
a)	$\chi_{\text{eff},i} = 0.0$	2.86°	7	$[-0.6, 0.6]$	<i>Medium</i>
b)	$\chi_{\text{eff},i} = -0.1$	2.92°	10	$[-0.7, 0.5]$	<i>High</i>
c)	$\chi_{\text{eff},i} = 0.1$	2.80°	9	$[-0.5, 0.7]$	<i>High</i>
d)	$\vec{S} \perp \vec{L}$	2.86°	6	$[-0.4, 0.4]$	<i>Medium</i>

spin induction in the non-spinning limit — for $p/M = 0.49$. For cases with nonzero effective spin, we make small adjustments to this angle to account for the influence of spin-orbit coupling (see Eq. (2.5.2)). This ensures that the induced spin-up is comparable to that of the non-spinning reference scenario. Once selected, the adjusted value of θ is consistently used across all simulations within the same effective spin configuration.

In the aligned-spin cases a)–c), we measure the change in χ_{eff} , $\delta\chi$, and the individual spin components $\chi_{1,2}^z$ at a final simulation time of $t_f = 285M$. This time is chosen to ensure that the spins have settled following the interaction. We then perform simple functional fits to capture trends in χ_{eff} and $\delta\chi$. For the configuration with perpendicular spins, we instead focus on the time evolution and final values of each individual spin component.

To simplify the analysis of these results we will just show the plots involving the data points

and their best-fits. The details of the parameters are not especially relevant for this TFM, but can be found in [72].

3.2.1. Spins parallel to $\vec{L}$

In Fig. 11, we display the induced changes in χ_{eff} (left panel) and $\delta\chi$ (right panel) for three distinct configurations in which the black holes (BHs) spin along the orbital angular momentum vector $\vec{L}$, labeled a)–c). The horizontal axis represents the z -component of the primary BH’s initial spin, shifted by the initial χ_{eff} so that the three cases are plotted around a common origin. Due to the inherent symmetry in the system for equal-mass black holes, it is sufficient to simulate only one half of the parameter space: swapping the two BHs keeps χ_{eff} unchanged (even symmetry), while it reverses the sign of $\delta\chi$ (odd symmetry), as follows from Eqs. (3.2.1) and (3.2.2).

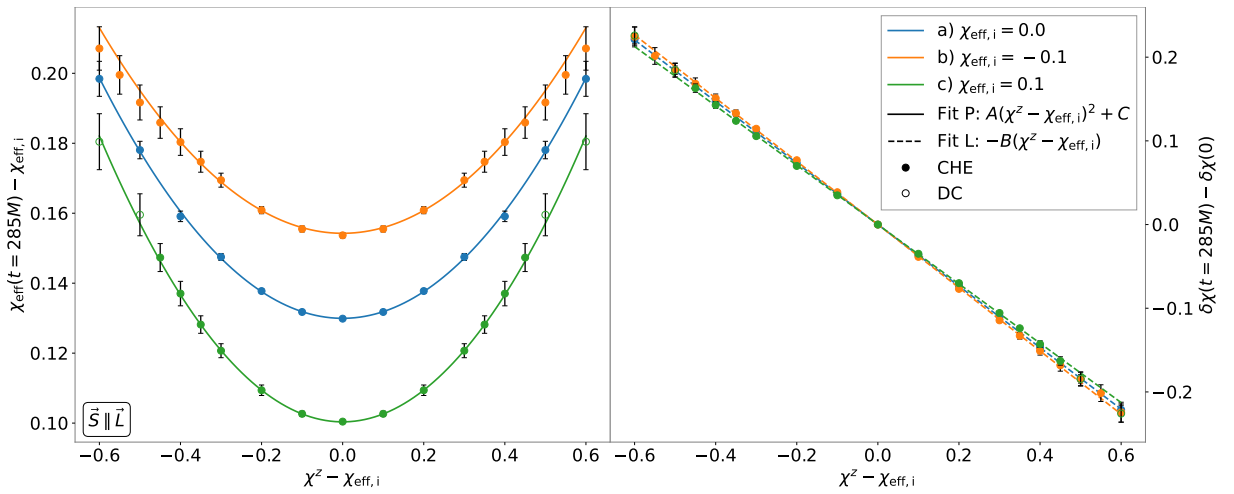


FIGURE 11: Evolution of the effective spin (left) and the weighted spin difference (right) measured at $t_f/M = 285$, along with their corresponding fits. The best-fit parameters (A , B , C) and their associated uncertainties are listed in the respective table in [72].

As seen in the left panel of Fig. 11, increasing spin asymmetry leads to an enhanced effective spin, driven by spin-orbit and spin-spin coupling effects — features absent in analyses of non-spinning captures [47]. The observed behavior is well described by a simple quadratic relation:

$$\chi_{\text{eff}}(t_f) - \chi_{\text{eff},i} = A(\chi^z - \chi_{\text{eff},i})^2 + C, \quad (3.2.5)$$

with the fitted parameters detailed in the analogous section in [72] for each configuration. Four data points corresponding to $\chi_{\text{eff},i}$ values mark (two) dynamical captures rather than scattering events; while excluded from the fitting process, their spin values at $t = t_f$ are still plotted for reference — as they have not yet reached the merger — and appear to roughly follow the established trends.

The remarkable accuracy of these fits, even for spins exceeding $\chi > 0.5$ at the start of the simulation, suggests the possibility of an underlying analytic explanation. Nevertheless, the curve corresponding to scenario b) deviates at the high-spin end. These discrepancies could stem from notable differences between *medium*- and *high*-resolution simulations in these regions, as reflected in the error bars, emphasizing the need for higher resolutions in more asymmetric

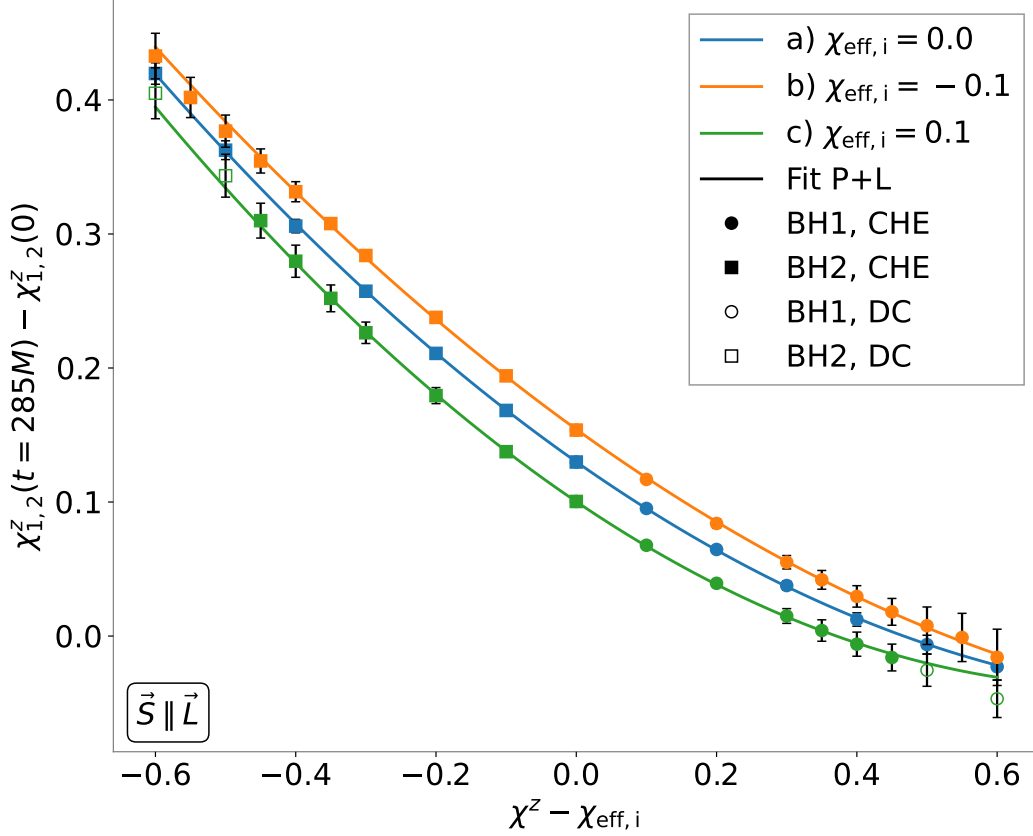


FIGURE 12: Spin-up values at $t_f/M = 285$ for each BH, computed from the fits shown in Fig. 11, using the coefficients from [72].

configurations.

Turning the analysis to the right panel of Fig. 11, the spin-weighted difference $\delta\chi$ (again relative to its initial value) decreases approximately linearly as χ^z increases. This trend is captured by the linear model:

$$\delta\chi(t_f) - \delta\chi(0) = -B(\chi^z - \chi_{\text{eff},i}), \quad (3.2.6)$$

with the corresponding fit coefficients (excluding dynamical captures) once again provided in [72].

Notably, increasing $\chi_{\text{eff},i}$ results in sharper parabolas and reduced spin-ups. Meanwhile, the values of B vary only slightly (within 7%), indicating that this parameter is largely independent of $\chi_{\text{eff},i}$.

By applying Eqs. (3.2.5) and (3.2.6) to the original definitions of χ_1^z and χ_2^z , we obtain the spin-up expressions for each BH:

$$\chi_1^z - \chi_1^z(0) = A(\chi^z - \chi_{\text{eff},i})^2 - B(\chi^z - \chi_{\text{eff},i}) + C, \quad (3.2.7)$$

$$\chi_2^z - \chi_2^z(0) = A(\chi^z - \chi_{\text{eff},i})^2 + B(\chi^z - \chi_{\text{eff},i}) + C. \quad (3.2.8)$$

These results are visualized in Fig. 12, showing how the spin-up for each BH diverges from the symmetric case as the initial spin asymmetry grows.

This behavior aligns with the theoretical expectations presented in Sec. 2.5., where we showed that the initially slower-spinning BH experiences a stronger spin-up at leading order. At second order, spin-spin interactions increase the spin-up of both BHs with rising spin-asymmetry, producing the observed parabolic pattern.

Interestingly, certain high-spin cases in Fig. 12 exhibit negative spin-up values. This suggests that in these scenarios, the influence of the initially negative-spin BH is dominant, outweighing other interaction terms in the system's dynamics. Specifically, this points to the significant role played by spin-spin interactions in shaping the spin evolution during close hyperbolic encounters.

3.2.2. Spins orthogonal to $\vec{L}$

We conclude our numerical exploration with a suite of simulations in which the initial BH spins are orthogonal to the orbital angular momentum vector, $\vec{L}$ (scenario d)). The initial spin configuration is chosen such that $\chi_1^x(0) = -\chi_2^x(0) = \chi^x$, with all other components set to zero. In relation to the symmetries of the system, interchanging the BH spins preserves the z -component but reverses the signs of the x and y components.

In Fig. 13, we present the spin-up of each spin component. In the left panel, we observe that the spin-up along the z -axis exhibits a parabolic dependence on the initial spin, suggesting that spin-spin interactions may be responsible for this behavior. Indeed, Eq. (2.5.3) shows a coupling in the interaction for spinning BHs with spins orthogonal to $\vec{L}$. Furthermore, since the

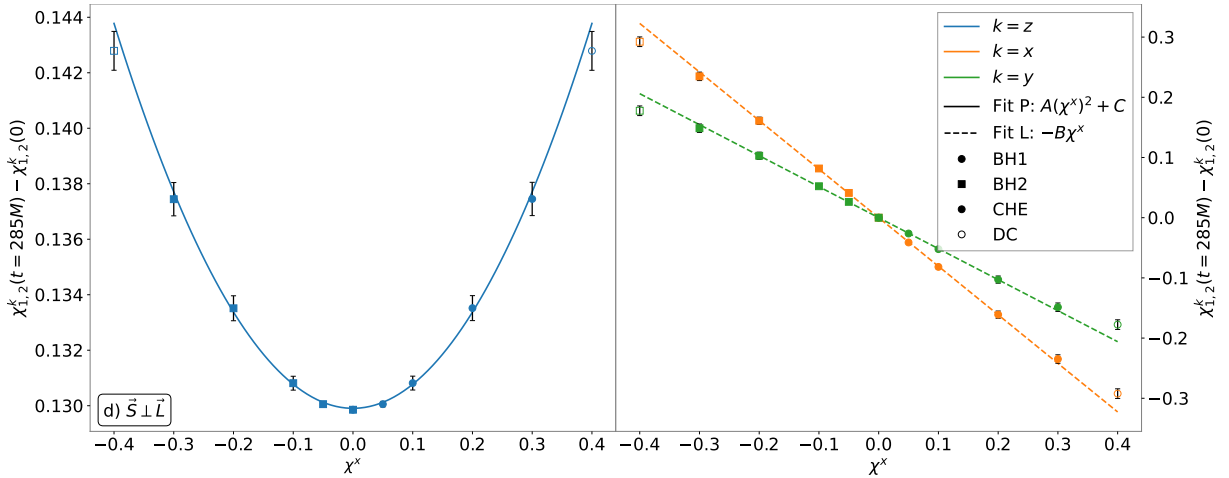


FIGURE 13: Spin-up in the z direction (left) and x , y directions (right). The fitted coefficients (A , B , C) along with their uncertainty are shown in [72].

spin-orbit term (Eq. (2.5.2)) cancels out initially, the spin-spin interaction (Eq. (2.5.3)) becomes more relevant. This Hamiltonian is more attractive for larger values of $|\chi^x|$. Additionally, as seen in scenarios a)–c), the spin-up of the z -component is enhanced when the initial spin difference between the two BHs increases.

In the right panel of Fig. 13, we observe that the spin-up in the x and y directions depends approximately linearly on χ^x . This behavior can be understood through the concept of spin precession, defined as a time evolution of the form

$$\frac{d\vec{S}}{dt} = \vec{\Omega} \times \vec{S}, \quad (3.2.9)$$

where $\vec{\Omega}$ is a precession vector that, due to the system's symmetry, is the same for both BHs and is aligned with $\vec{L}$.

Figure 14 illustrates the 3D trajectory of one simulation in scenario d), along with the corresponding spin vectors, highlighting the effects of precession. The spin vectors evolve along the direction of $\vec{L}$, in agreement with Eq. (3.2.9). During this precession, the BHs acquire a boost in the positive z -direction. Swapping the spins results in a motion toward negative z , though the orientation of the orbital plane remains unchanged, as the total spin is zero. Consequently, no precession is induced on $\vec{L}$.

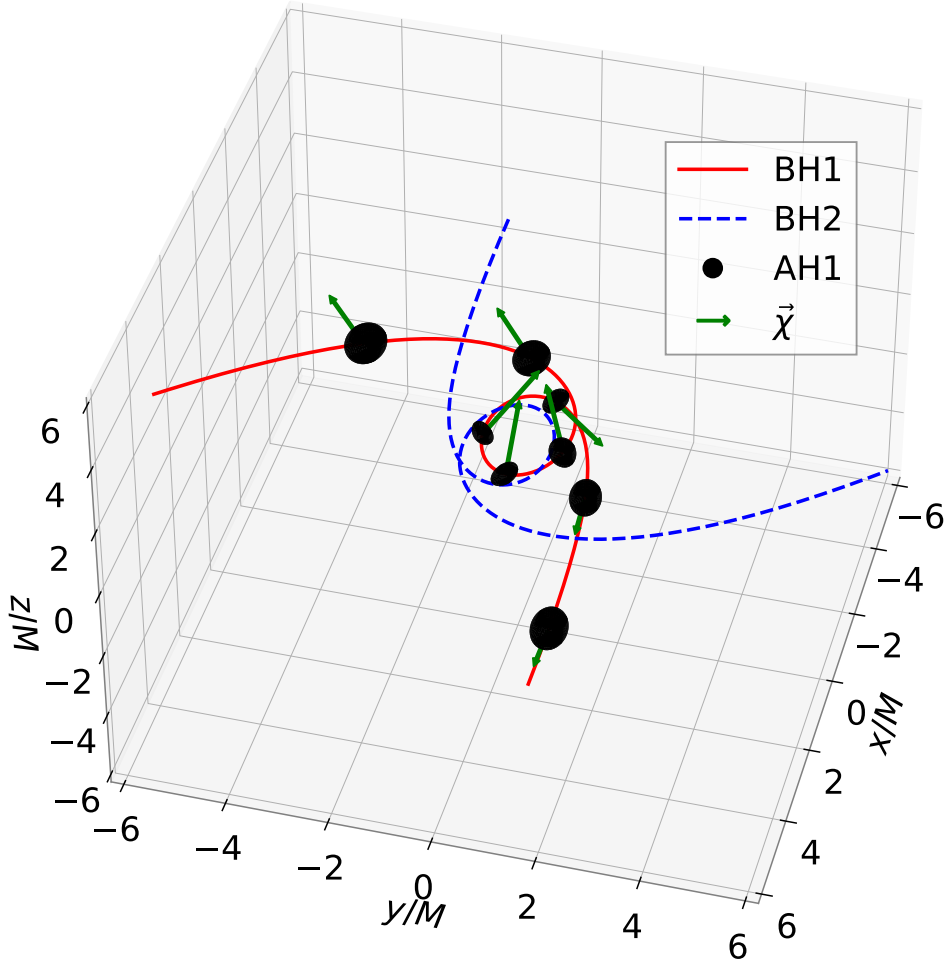


FIGURE 14: Visualization of the precession of BH spins in a CHE, where the initial spins point along the x -axis in opposite directions.

The evolution of the spin components can be approximately described by the following expressions:

$$\chi_{1,2}^k(t_f) - \chi_{1,2}^k(0) = \mp B^k \chi^x, \quad k = x, y; \quad (3.2.10)$$

$$\chi_{1,2}^z(t_f) - \chi_{1,2}^z(0) = A(\chi^x)^2 + C. \quad (3.2.11)$$

The fitted coefficients are reported in the analogous subsection of [72], including the 68% confidence-level uncertainties. As in the previous subsection, we exclude the dynamical capture

case ($\chi^x = \pm 0.4$), which deviates significantly from the linear trends observed in the remaining data.

3.2.3. Trajectories

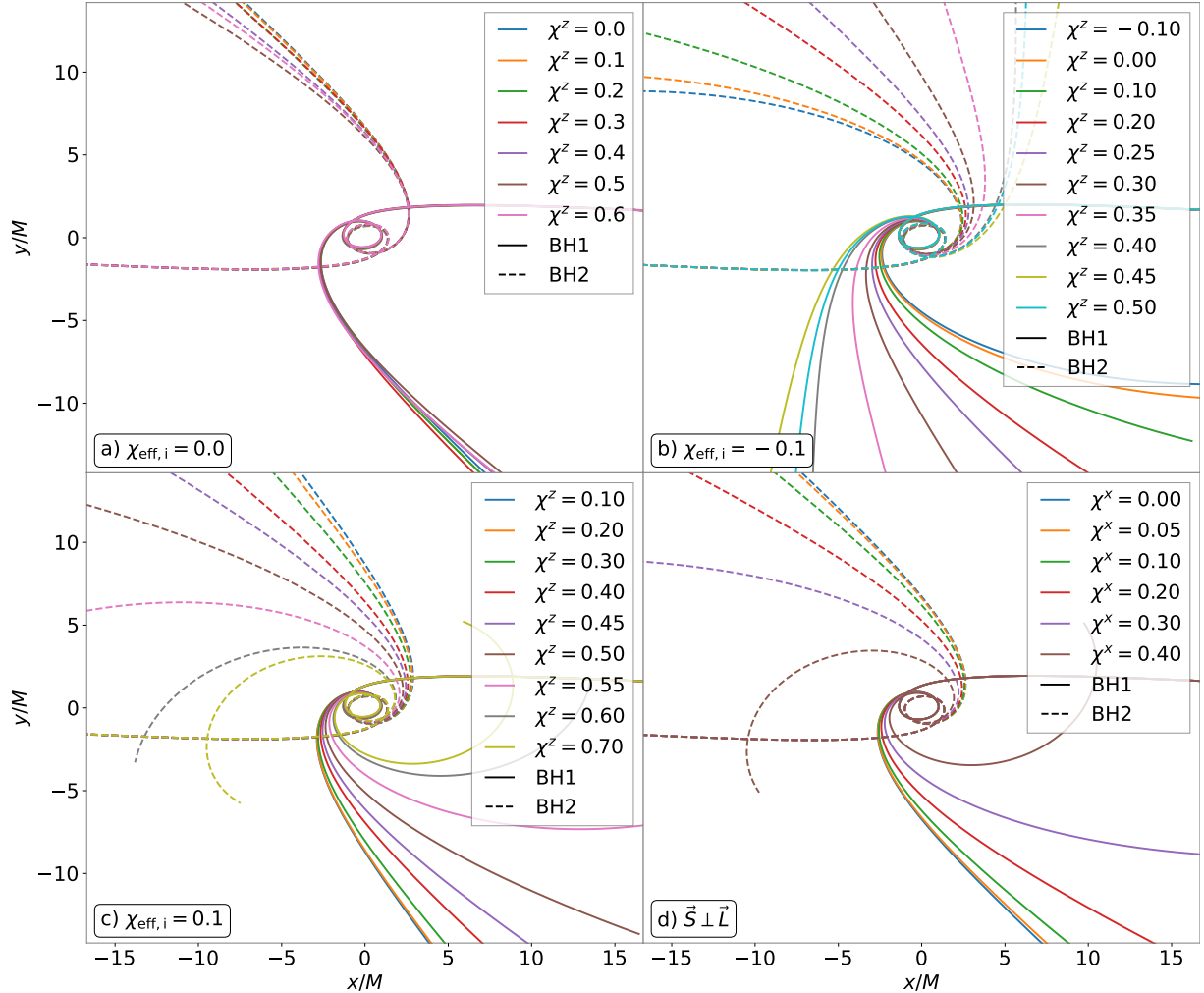


FIGURE 15: Trajectories of each CHE configuration discussed in Sec. 3.2.. For negative effective spins, larger spin asymmetries appear to enhance repulsive behavior, as illustrated in scenario b). In contrast, positive effective spins (scenario c)) exhibit greater attraction for increasing spin differences. Cases with zero effective spin (scenario a)) show minimal variations in scattering angle, while varying spins along the x -direction (scenario d)) quickly lead to dynamical capture.

To gain a deeper understanding of the simulation dynamics, we display the BH trajectories in Fig. 15. These reveal how the scattering angles evolve with varying initial spin configurations, in some cases culminating in dynamical capture. In scenario a), the trajectories are relatively insensitive to changes in spin, while scenarios b), c), and d) exhibit more pronounced variations in scattering angle.

In particular, scenario b) becomes increasingly repulsive as the spin difference grows, whereas scenarios c) and d) show the opposite trend, with stronger attraction under greater spin asymmetry. This behavior is not fully explained by the spin-spin coupling term in Eq. (2.5.3), indicating that additional dynamical effects may be at play, motivating further investigation.

Scenario d) in particular leads to notably close encounters when $|\chi^x| \gtrsim 0.3$, as seen in Fig. 15.

This could be attributed to frame-dragging effects: opposing spins aligned along the x -axis induce a mutual boost in the z -direction, drawing the BHs closer — an effect previously noted in Fig. 14. Simulations with larger χ^x values were excluded from the analysis, as they resulted in dynamical captures occurring before $t/M = 300$. This attractive tendency is also consistent with the behavior predicted by the spin-spin interaction in Eq. (2.5.3).

Finally, it should be noted that we did not perform a detailed analysis of the scattering angles due to uncertainties that reach approximately 8° . Higher-resolution simulations would be necessary for a more accurate and comprehensive study of these deflection angles.

4. Dynamical captures

In the previous chapter, we have shown some relevant effects that lead to spin induction, considering initially spinning black holes. Now, we are interested in exploring the properties of DCs for initially non-spinning BHs: from the GW emission patterns for constructing a future waveform model regarding CHEs and DCs, to the study of the intermediate and final masses and spins of the interacting and formed BHs.

As mentioned before, CHEs have gained more and more attention due to their possible implications for the evolution and dynamics of PBHs in dense clusters. Consequently, since a DC is a limit where the hyperbolic encounter emits enough energy and angular momentum to bind the orbits and lead to a merger, these dynamical scenarios will gain attention in the coming years [5, 58], due to their expected relevance — along with pure scattering events — in dense environments of BHs. Both of these phenomena will provide the necessary effects and mechanisms that will be indispensable for building a precise and efficient N-body simulation regarding dense clusters of black holes, with which we will determine mass and spin distribution, as well as detecting the GWs coming from sources within these dense environments.

However, their detectability is compromised in LVK detectors, as DCs require long measurement times and high precision. Nevertheless, as detector sensitivity improves, we might be able to distinguish a DC in some of the runs of the LVK collaboration.

In this chapter, we go back to initially non-spinning black holes with equal masses ($q = 1$) for simplicity, and the inclusion of spin and different masses in DCs will be left for future works. A justification to use initially non-spinning black holes is that PBHs are expected to be formed with low spins [24], where their spins change over time with subsequent interactions (mainly CHEs), as seen in the previous chapter. Given that previous works have been done regarding similar initial BH configurations [47, 59, 72] with identical refinement factors and grid properties (see Sec. 2.2.), we infer that the differences between what we have previously called *medium* and *high* resolutions are always smaller than 0.6%, and so we will not include an error analysis section here. This is especially true because the maximum induced spins from our non-spinning black holes are $\chi \sim 0.14$ which is essentially almost identical to one of the cases in [72], which also has fewer differences between resolutions. However, this was cross-checked, and we concluded that using *medium* resolution provides very accurate results.

4.1. Choice of initial conditions

We perform simulations with the configuration in Fig. 2 with values of momenta in the range $p/M = 0.095 - 0.49$, incidence angles with values $\theta = 6.36^\circ - 2.83^\circ$ and initially without spins. We show, in fact, a table with the number of simulations for each value of selected linear momenta and the range of impact parameters and incidence angles Tab. 2.

In the aforementioned table, we see that the ranges of incidence angles and of impact parameters that lead to capture are bigger, the smaller the BH's linear momenta. This seems logical, as it requires a stronger interaction (which can be provided by a smaller incidence angle) to capture a more rapidly moving BH.

In the following section, we will study all the properties of the BHs during their interaction

TABLE 2: Considered values of p/M along with their respective ranges of θ and the fitted ranges of impact parameters, b/M . The value N_S represents the number of simulations that have been computed for the given scenario.

p/M	θ (deg)	b/M	N_S
0.095	6.21 – 6.35	10.84 – 11.09	9
0.1225	5.04 – 5.14°	8.81 – 8.97	10
0.245	3.29 – 3.32	5.75 – 5.80	12
0.3675	2.90 – 2.93	5.06 – 5.11	10
0.49	2.83 – 2.86	4.94 – 4.98	9

and after the merger, as well as estimating the characteristics and behavior of the GW emissions — introduced and described back in 2.. In the case of mass, irreducible mass and spin we will be interested in the instants where these parameters have stabilized, where the curves flatten for long time intervals (see Fig. 7). In the case of the GW emission we will of course be interested in the regions where there is prominence of the Weyl scalar, analyzing the amplitude of each emission, their frequency, phase, etc⁸.

4.2. Study of the Weyl scalar

First, we aim to model the behavior of the Weyl scalar for each simulation. For this, we start with a simple fitting procedure:

- The first emission — the one for the CHE — will be described with a *sin-Gaussian* function:

$$\Psi_4^{\text{CHE}} = A_{\text{CHE}} \exp \left\{ -\frac{(t - t_0^{\text{CHE}})^2}{2\sigma_{\text{CHE}}^2} \right\} \sin(\omega(t - t_0^{\text{CHE}}) + \phi_{\text{CHE}}), \quad (4.2.1)$$

for both the imaginary and real parts.

- The second emission — the one for the merger and ringdown — will be divided into two portions: the first one will have the same expression as Eq. (4.2.1) — changing the labels “CHE” with “M” (referring to merger) —, while the second will be modeled with the first ringdown mode ($n = 0$), which according to [68] takes the form:

$$\Psi_4^{\mathcal{R}} = \sum_n A_n^{\mathcal{R}} \sin(\omega_n^{\mathcal{R}}(t - t_a) + \phi_n^{\mathcal{R}}) e^{-(t-t_a)/\tau_n^{\mathcal{R}}}, \quad (4.2.2)$$

where t_a is the chosen cutoff value which separates merger trend from ringdown trend. These decaying modes appear because the final BH that forms during the merger settles into a stable, final state by radiating away its distortions through GWs.

In order to add consistency and improve the accuracy of the fitted parameters, we will first fit the capture amplitude — with a regular *Gaussian* function — $\Psi_4 = \sqrt{\Psi_{4,R}^2 + \Psi_{4,I}^2}$ to achieve

⁸Analogously to the previous chapter, we will use the term “case” for an individual data point (pair of initial conditions $\{p/M, \theta\}$), and the term “scenario” for the ensemble of cases with a common p/M .

common values of A_{CHE} , t_0^{CHE} and σ_{CHE} ; with the same being for the merger portion of the Weyl scalar. This assumes two things: the first is that the measured frequency is the same for both the imaginary and real parts, and the second is that the difference between the phases of both the imaginary and real parts will be $\pi/2$. This is expected for a well-behaved GW.

In fact, once we have fitted the parameters relative to the amplitude, we then fit simultaneously the real and imaginary parts, forcing both to find a common best-fit value for the frequency, while leaving the phase as a free variable, and then checking how well do they fulfill the condition that both phases must have a difference of $\pi/2$.

On the other hand, the fitting procedure for the ringdown portion of the Weyl scalar is done in the same manner as for the CHE and merger, but with its decaying mode. This can be done because we have chosen the first mode; if we were to choose two or more, the analysis would be more complicated, as the two components chosen to fit the peaks would not allow for a simple fitting of first the amplitude-related parameters and then the frequencies and phases of the Weyl scalar from the real and imaginary parts.

We plot an example (Fig. 16) of the performance of these fits with a previously shown example (Figs. 6 and 7). Note that we plot the imaginary and real parts separately, along with their fitted counterparts, with the procedure explained earlier. It can be seen here that the performance of these fits is quite good, especially for the ringdown — at a first glance. One can observe, as well, that the main differences with respect to the simulation output are found in the tails of the *sin-Gaussian* fits. This is probably due to the frequency of the GW not being actually constant, due to Kepler’s law — knowing that there is a fixed relation of the orbital and GW frequency of $\omega = 2\Omega$ [54] —, as the distance of the BHs is not constant throughout each emission.

We will determine effective trends for each of the fitted parameters in terms of the incidence angle (θ) to help us build a future DC waveform model. However, first we first carry out the mentioned *sin-Gaussian* fits and obtain an important measure for simplifying our analysis. With the parameters t_0^{CHE} and t_0^{M} (from Eq. (4.2.1)), we obtain the effective distance from one peak to another $\Delta t \equiv t_0^{\text{M}} - t_0^{\text{CHE}}$. This is done because we know that when we increase the angles, we get to a threshold value that separates CHEs and DCs [59]. In terms of the distance of the peaks, what this implies is that when we reach this threshold value, $\Delta t \rightarrow \infty$ — as pure CHEs do not ever reach a merger —, so we expect that there is a singularity in the trend of the distance of the peaks, when plotted against the incidence angle, which is exactly what we observe in the following. The utility of obtaining this threshold angle (θ_0) — in addition to calculating where the separation between DC and CHE is found — is that we can “normalize” our axis when obtaining the desired trends for each of the parameters, having more compact plots considering the disparity of values observed in our ranges of θ (and b/M), as already shown in Tab. 2.

The distances between peaks ($\Delta t/M \equiv t_0^{\text{M}} - t_0^{\text{CHE}}$) for each scenario, along with their parametrization (represented using a solid line) — captured by the following expression

$$\Delta t/M = \frac{A}{1 - \theta/\theta_0} + B \quad (4.2.3)$$

—, is plotted in Fig. 17, where the dots represent the obtained data with the fits of both *sin-Gaussian* functions, as explained earlier. The fitted parameters A , B , θ_0 for each scenario can be found in Tab. 3.

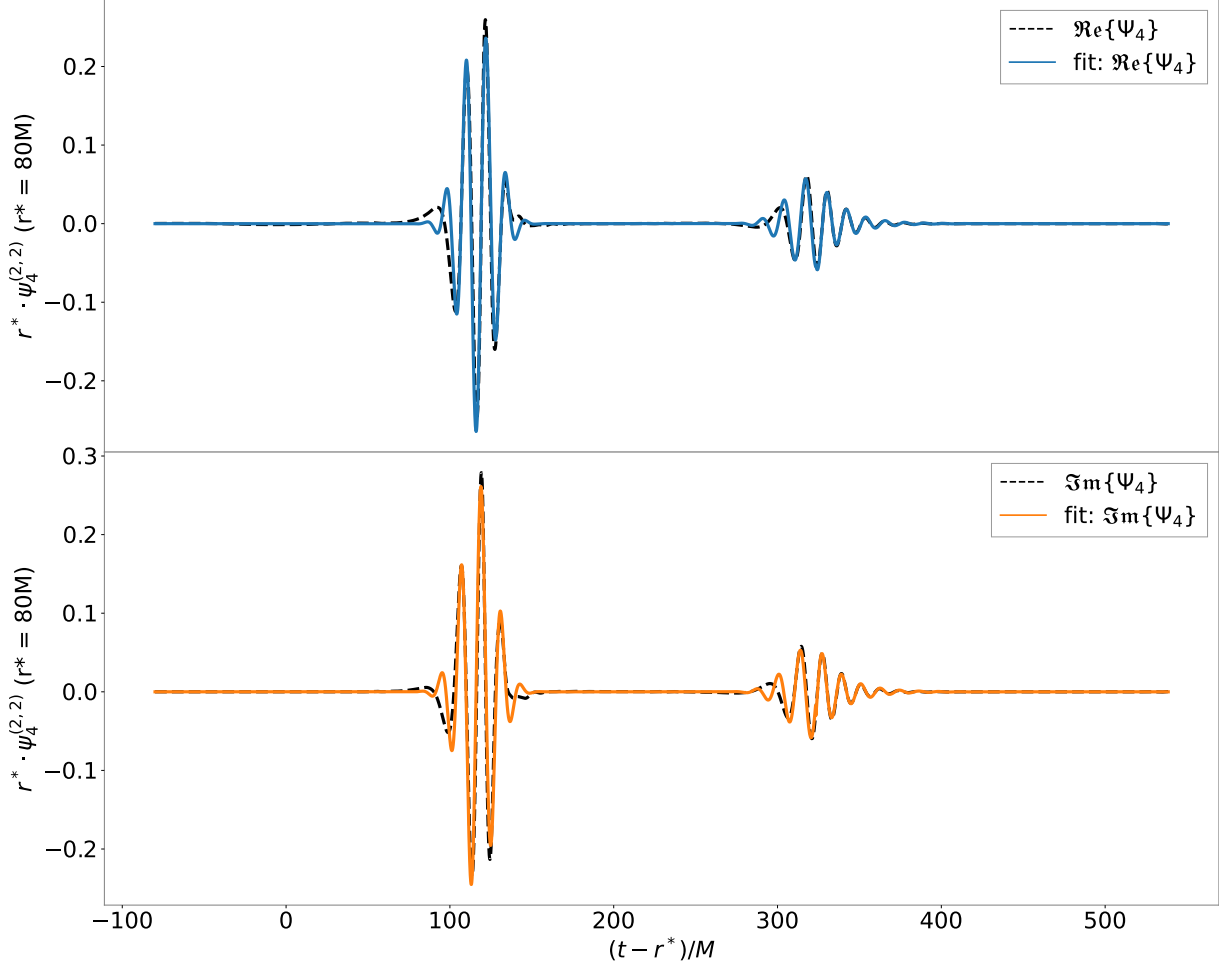


FIGURE 16: Fits of the real (blue) and imaginary (orange) parts — upper and lower panel respectively — along with the simulated signal (black, dashed) of the rescaled Weyl scalar at a detector at a distance $r^* = 80M$ from the CM. The initial conditions that give this Weyl scalar are the ones of the example case described in Sec. 2.4..

TABLE 3: Fitted parameters according to Eq. (4.2.3). We also show in parentheses their respective 68% confidence-level uncertainties for the last digit(s). The obtained curves are plotted alongside the fitted data points in Fig. 17.

p/M	θ_0 (deg)	A	B
0.095	6.40401(172)	14.08(63)	-358.9(224)
0.1225	5.16308(75)	7.600(291)	-285.5(144)
0.245	3.32992(12)	1.923(46)	-147.4(60)
0.3675	2.93060(4)	1.337(21)	-108.0(37)
0.49	2.85889(4)	1.145(19)	-91.42(360)

We can now obtain the trend that θ_0 follows with respect to the initial p/M . In a previous section (Sec. 2.4.), we have mentioned that larger values of p/M lead to lower values of θ_0 ; as the faster a BH moves, the closer it has to be to the other one for the encounter to be sufficiently strong to produce an effect of capture. We have, however, detected that highly relativistic encounters seem to break this trend, as the threshold angle increases again for $p/M \gtrsim 0.49$. The

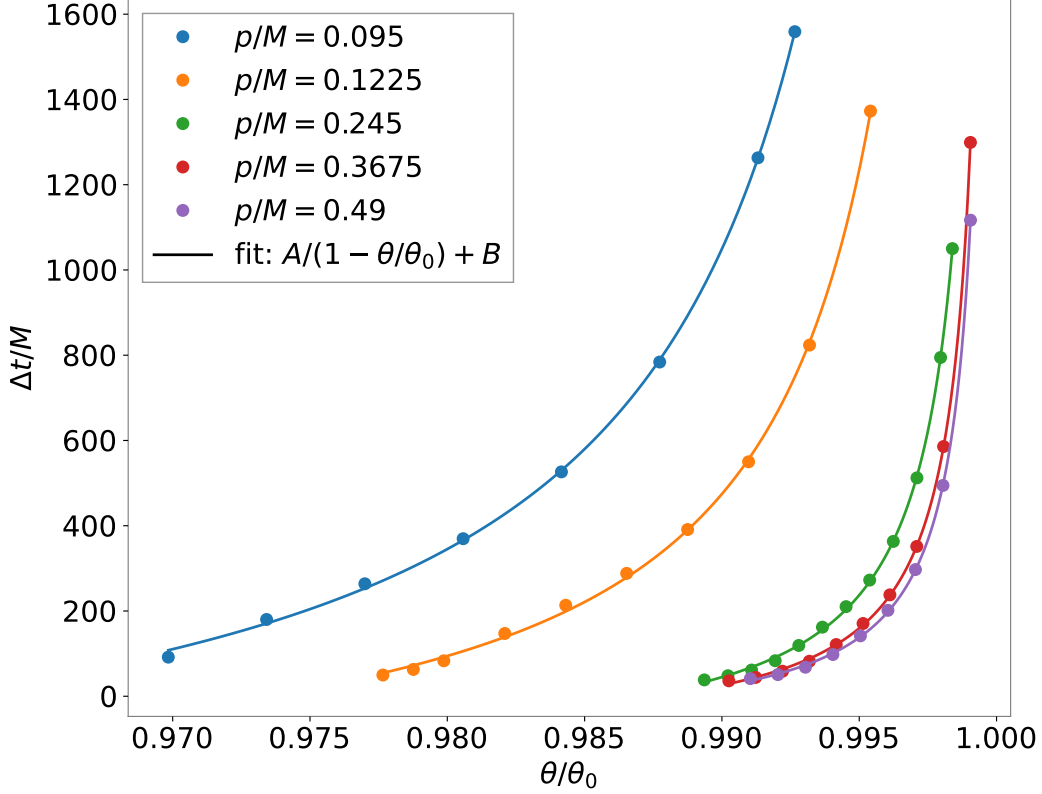


FIGURE 17: $\Delta t = t_0^M - t_0^{\text{CHE}}$ with respect to θ/θ_0 the data points are obtained with Eq. and the lines are fits according to the expression in Eq. (4.2.3).

obtained values for θ_0 in Tab. 3 are plotted with respect to their initial p/M in Fig. 18, and we include an estimated value of θ_0 for $p/M = 0.75$ — which emphasizes the breaking of the trend. We have not included this scenario ($p/M = 0.75$) in our study of waveform fitting because — as of the time this Thesis was written — we have not simulated enough cases for a detailed analysis, but we have estimated the threshold value in the incidence angle (by trial and error) to be at around $\theta_0 \approx 3.1^\circ$.

The equation that has been observed as best-fit for the trend followed by $\theta_0(p)$ is

$$\theta_0(p) = \frac{A'}{p} + B' + C'p, \quad (4.2.4)$$

with $A'M = 0.529(6)$, $B' = 0.580(62)$ and $C'/M = 2.43(9)$, with the values in parentheses representing 1σ uncertainties on the last digit(s).

We expect that the highly relativistic cases that currently seem to follow a positive slope given by the C coefficient, saturate up to a certain value of θ_0 , which is reason for further investigation, as with the current number of values it is impossible to properly quantify this hypothetical saturation value.

In general, it is not clear why this value increases. It may be due to spin-orbit and spin-spin effects that become considerably prominent at these values of p/M , as the spin-induction values become non-negligible for $p/M > 0.3675$. In a first approximation, one can see that according to Eq. (2.5.3) two parallel spins create a negative contribution in the Hamiltonian, increasing the attraction between BHs, so a noticeable spin-induction along $\vec{L}$ for both BHs could be increasing

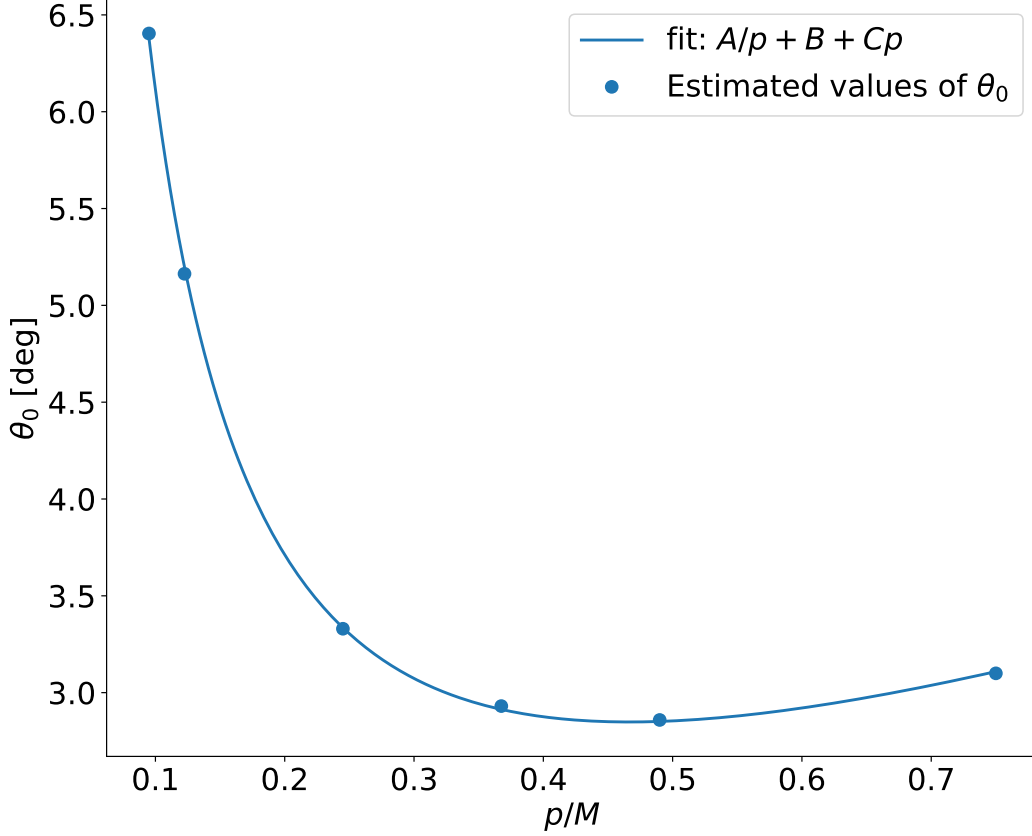


FIGURE 18: Estimated values for θ_0 from Tab. 3, plus an additional case for $p/M = 0.75$, which was approximated by running various simulations (both pure CHE and very head-on encounters) and estimating where the threshold value lies. These values are plotted over a fit encapsulated in Eq. (4.2.4).

the threshold angle θ_0 (as this additional “attractive” contribution would ease entering the DC regime). There are other effects which may affect, such as the horizon absorption, and their interplay would help explain this phenomenon. However, as of now, we do not have a proper physical explanation, which motivates future research on this matter.

4.2.1. Analysis and model of the CHE emission

We represent the data points obtained from the fits performed on the first emission (as seen in Fig. 16). Depending on the observed nature of the trend of each parameter, we have fitted them to either a linear function

$$y = a(\theta/\theta_0) + b, \quad (4.2.5)$$

or to an exponential decay of the form

$$y' = a \exp\{-b(1 - \theta/\theta_0)\} + c, \quad (4.2.6)$$

where $y \equiv \{A_{\text{CHE}}, \omega_{\text{CHE}}, t_0^{\text{CHE}}\}$ and $y' \equiv \{\sigma_{\text{CHE}}, \phi_{\text{CHE,R}}, \phi_{\text{CHE,I}}\}$, as seen in the following.

As seen in Fig. 19, the emission regarding the CHE fits very well with the proposed expressions, and we obtain well-behaved trends for each of the parameters, clearing the path for a waveform model for CHEs. There are, however, some parameters ($A_{\text{CHE}}, \omega_{\text{CHE}}$) which at the shown scale

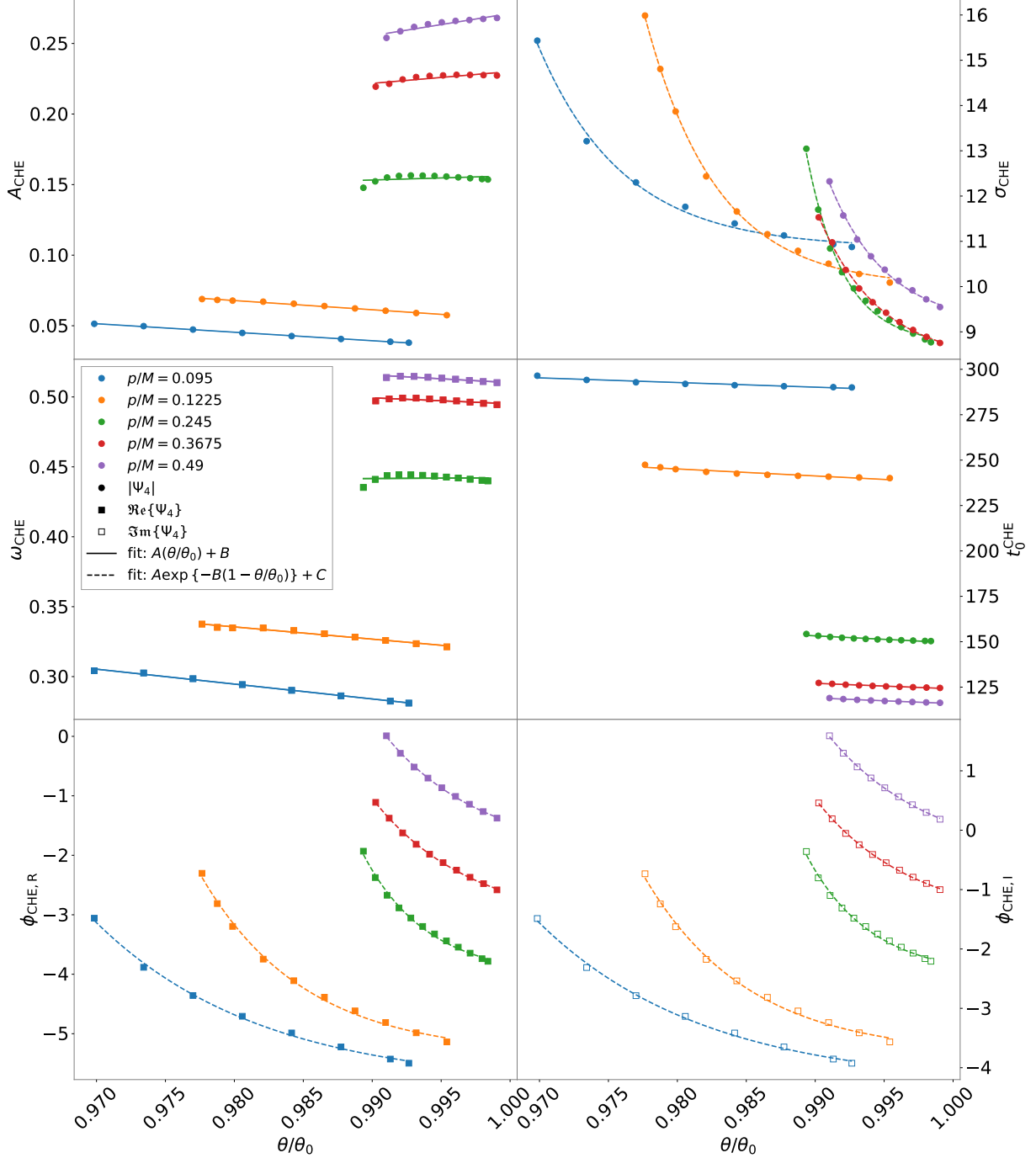


FIGURE 19: Estimated parameters from the fitted CHE portions of the Weyl scalars for all values of p/M . Two type of functions are used to fit these parameters: a linear trend, represented with a solid line (Eq. (4.2.5)), and an exponential decay, represented with a dashed line (Eq. (4.2.6)). We also show how we compute these values: the ones represented with a solid dot are calculated by fitting the modulus of the Weyl scalar, while the ones with solid and empty squares are computed by fitting the real and imaginary parts, respectively. The fitted frequencies overlap for both components because we have imposed then to find a common best-fit. In the Annex of this text we include a table (Tab. 4) of the values that the coefficients in their respective fits yield.

do fit the linear trend but may have non-linear dependencies, as the data points differ from the linear trend in an apparently continuous and well-behaved manner. In fact, we can give physical sense to the trend that these two parameters seem to follow. We observe that these two

parameters seem to first grow with θ/θ_0 and then — after a critical point — acquire a negative slope (the purple points may be an exception to this phenomenon due to highly relativistic effects, although a change in the behavior is also observed), decreasing its magnitude. This may be because the CHE produces a GW emission — weaker for higher incidence angles — which alters the orbit from a hyperbolic to a short-lived eccentric orbit which then falls into a merger; however, if the incidence angle is too small the initial trajectory will be almost head-on, reducing the need of such a strong GW emission to bind the orbit.

In the Annex of this text, we write a detailed table (Tab. 4) that shows the values and uncertainties of the coefficients arising from the aforementioned fits.

With this, the characterization of the Weyl scalar in terms of known functions works in a very satisfactory way. In the future, however, we intend to better fit the differences that appear at the tails of this distribution; perhaps checking how a time-dependent frequency performs.

Additionally, one thing to note is that, given the current sensitivity of LVK detectors, the waveform model that may come from this study could be enough to carry out a parameter estimation of CHEs, or at least signal detection of these events, which as of now have not been detected.

4.2.2. Analysis and model of the merger-ringdown emission

As hinted earlier, the study of the Weyl scalar at the merger-ringdown phase must be divided into two portions (ensuring continuity and differentiability between both), according to Eqs. (4.2.1) and (4.2.2). We will see that the trends here are more complicated to analyze, due to the fact that we obtain more differences between imaginary and real parts than in the previous subsection.

The time at which we separate the merger and ringdown phases has been chosen at $t_a = t_0 + 5M$ (according to Eq. (4.2.2)), where t_0 is the time at which the maximum of the second emission is found (generally $t_0 \approx t_0^M$, but not identical).

Merger

In the Annex we show the obtained parameters for the merger portion of the last emission for each of the cases, in a similar fashion to the study of the CHE emission in the previous subsection. If one checks the corresponding figure (Fig. 23) one will see that there are no clear trends that these parameters follow with respect to θ/θ_0 , and so, we have not considered necessary to calculate any fits for these estimated parameters, as was actually done in Tab. 4 for the CHE estimated parameters.

There are two reasons to understand the difference between the accuracy of the CHE estimation compared to the “inaccuracy” of the merger emission:

- Firstly, the order of magnitude of the estimated parameters for all values of p/M (scenarios) are of about the same order. If we look at Fig. 19, one can see that the data points are nicely distributed over the plot, while for the merger emission they are all clustered around almost the same order of magnitude for each parameter. This shows that after the CHE, for any value of p/M and θ/θ_0 , we are left with a bound system which merges emitting similar amounts of energy and angular momentum.
- Secondly, and most importantly, the fact that we separate the emission in two phases may

be a very naïve ansatz. On the one hand, is not clear when the ringdown starts, as it begins at slightly different times for the real and imaginary parts, so it is not entirely accurate to say that the ringdown starts at a fixed time for both components of the Weyl scalar. On the other hand, we have fitted the *sin-Gaussian* function to the whole merger-ringdown emission, which may be hindering the quality of these fits, as it tries to fit the ringdown portion as well. One may wonder why not just use the first portion of the peak, but we tried this and the results were even worse, because very non-linear effects may be at play, as well as the fact that we only would have one portion of the peak to estimate the properties of the proposed *sin-Gaussian* evolution.

Nevertheless, the obtained parameters perform decently. The fit of the merger amplitude performs especially badly for DCs where both emissions — the CHE and merger-ringdown phases — are close to each other ($\Delta t/M \lesssim 100$), almost mixing with each other. This occurs because, in those cases, both emissions fuse together at the tails, creating very non-linear trends, which cannot be described by the simple distributions we use here (an example of this behavior can be found in the Annex in Fig. 25). However, one parameter that performs really well — as already hinted since it was used to compute the values in Fig. 17 for $\Delta t/M$ — is t_0^M , where as seen in the figure of the Annex (Fig. 23), the sums of the previous fits $\Delta t + t_{\text{CHE}}$ plotted over the data points of t_0^M work extremely well.

Ringdown

Now, we carry out the fit for the ringdown portion of the last emission and observe that once again the parameters cluster around the same order of magnitude in each of the plots (as seen in the Annex, Fig. 24). Here, we proceed in the same way as before: fitting first the amplitude, fixing its values, and then fitting each real and imaginary parts imposing that they find a common best-fit of the frequency and leaving the phase of both components freely.

Although the results for the ringdown were particularly good with regard to following the time evolution of the Weyl scalar (see Fig. 16), we have seen that the estimated parameters for the ringdown in some cases oscillated, not giving clear trends in these parameters, especially for the lower values of θ/θ_0 in each scenario (as can be seen in Fig. 24, in the Annex). In the near future we will attempt to improve the quality of these fits, maybe delaying even further the start of the cutoff time where the ringdown starts while also improving the estimation in the merger portion of the peak.

As mentioned already, the immediate goal we have for this project is to find a clearer way to describe the last emission, even if the current study is good enough to have an idea of the orders of magnitude of the parameters involved in the merger-ringdown evolution.

4.3. Final BH properties

Once we have analyzed the properties on the measured Weyl scalar, we are interested in studying the intrinsic properties of the BHs: spin and mass, along with other relevant variables that arise from these other two. These quantities are obtained with the QUASILocalMEASURES thorn, as stated in Subsec. 2.1.1..

All the variables regarding the final state of the BH must be chosen after we reach the relaxed state. This instant was chosen at a time of $140M$ after the merger for all cases.

4.3.1. Analysis of spin

In Fig. 20, on the left panel, we plot the spin of the final BH for the fixed time of $140M$ after the merger of each case. Here, we observe that the final spin of the BH varies in the range $\chi_f \in [0.67, 0.77]$. In this figure, one can see that spins decrease when increasing θ/θ_0 , then they reach a different minimum for each value of p/M , and then they start growing. This curious trend can be explained in a similar way to the slight deviations detected for the parameters $(A_{\text{CHE}}, \omega_{\text{CHE}})$ referenced in Subsec. 4.2.1.: before the minimum in χ_f we are in a regime close to head-on collision, in which not so much GW emission during the CHE is necessary to bind the orbit — so the energy and angular momentum losses are less significant for this regime —; then, we approach a critical point (in θ/θ_0) where this emission is maximal — meaning that the energy and angular momentum losses impact the final spin, reducing it —; finally, the angles keep increasing leading to a further (and weaker) encounter, hence less GW emission — which implies a greater final spin, as greater angles involve greater angular momentum, which then transforms into final BH spin.

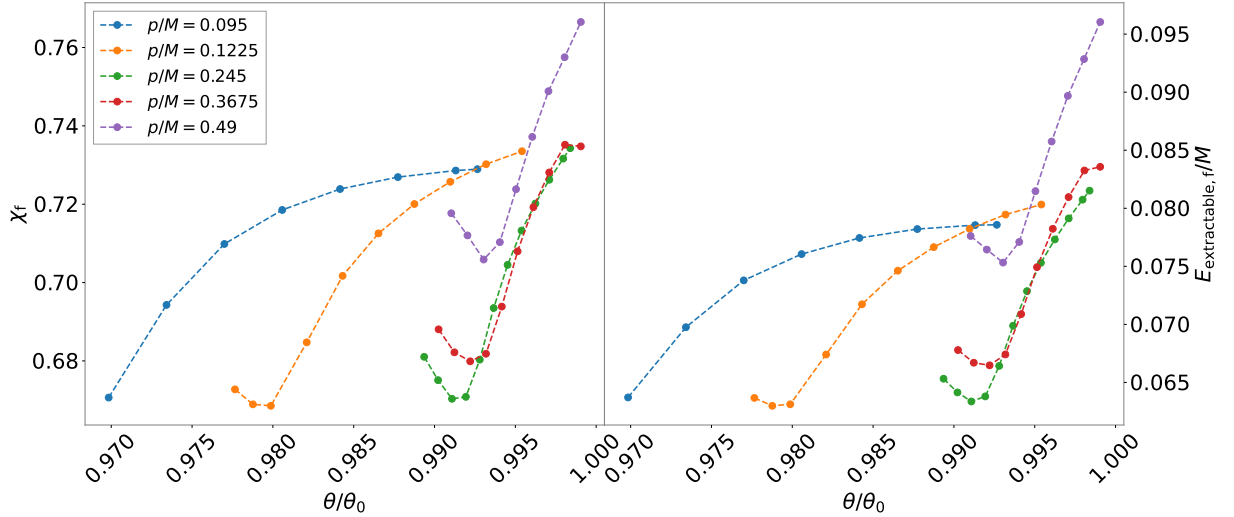


FIGURE 20: Spin (left panel) and extractable energy (right panel) of the final BH with respect to the normalized angle θ/θ_0 for each scenario.

The rightmost red point in Fig. 20 seems to break the growing tendency for its curve, but this is probably an unphysical effect of precision. Such high angles involve very long (simulation) times which accumulate significant numerical errors. This point will also be problematic for the mass study.

Additionally, one can observe that the two lower values of p/M yield similar trends in terms of their final spins. Then, the green and red curves have orders of magnitude similar to the blue and orange ones, but a significant positive slope after the critical point seems to appear. Finally, the purple curve (the one for $p/M = 0.49$) seems to differ the most from the rest. We suspect that these differences are due to the more prominent induced spins during the CHE for higher values of p/M , as stated in [47, 59]. Thus, while the final merger of BHs in the $p/M = 0.095 - 0.245$ scenarios occurs with basically no induced spins after the CHE (leaving “only” the remnant angular momentum to contribute to χ_f), the scenario $p/M = 0.49$ has significant induced spins ($\chi \sim 0.14$) before the merger, leading to a higher BH spin (as both BH spins and the remnant angular momentum contribute). The scenario $p/M = 0.3675$ lies somewhere in the middle, as the induced spins are of the order $\chi \sim 0.07$.

On the right panel of Fig. 20 we plot the extractable energy, which will be introduced in the next subsection. We plot it beside the final BH spins plot due to their similar trends, which will also be explained in the following.

4.3.2. Analysis of the BH masses

In Subsec. 1.3.4. we described the relation between ADM and irreducible masses (see Eq. (1.3.9)). We plot these quantities on both panels of Fig. 21, respectively.

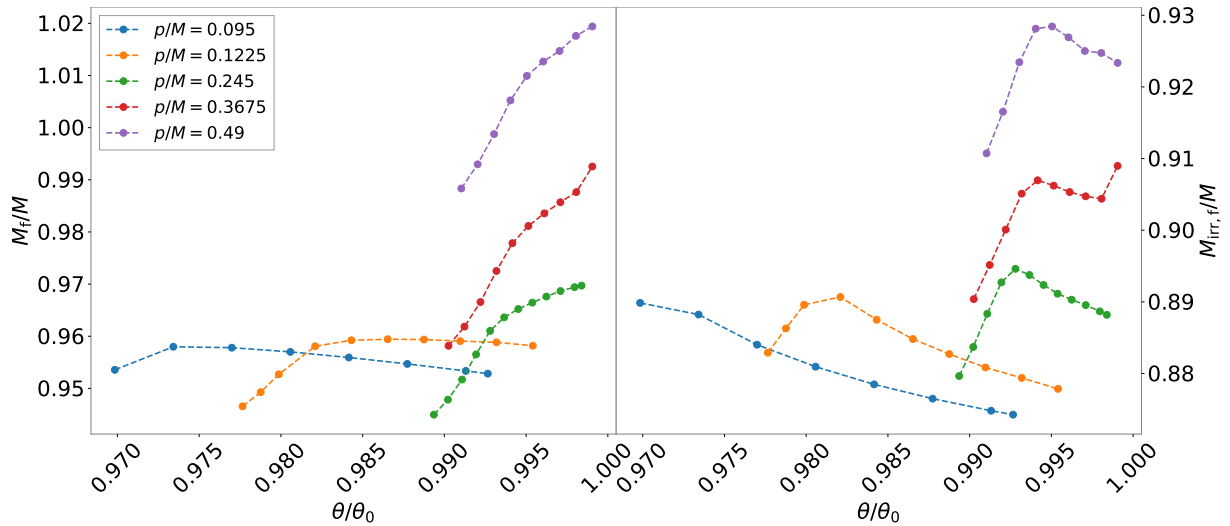


FIGURE 21: ADM (left panel) and irreducible (right panel) masses of the final BH with respect to the normalized angle θ/θ_0 for each scenario.

In the right panel of Fig. 20, we plotted the extractable energy (see the right panel of Fig. 20) — in terms of θ/θ_0 — from the final BH ($E_{\text{extractable}} \equiv M_f - M_{f,\text{irr}}$), which is the maximum energy that can be extracted from a Kerr BH through the Penrose process or BH mergers [74]. There, it is shown that the extractable energy follows very similar trends to the ones seen for χ_f on the left panel of that same plot, which hints that the extractable energy is directly affected by the contributions of the final spins.

In fact, if we Taylor expand around $\chi_f = 0.686$ (this value is chosen because it is the spin of the final BH in a quasicircular merger), the expression of the irreducible mass (obtained from Eq. (1.3.9))

$$M_{\text{irr},f} = M_f \sqrt{\frac{1 + \sqrt{1 - \chi_f^2}}{2}}, \quad (4.3.1)$$

yields the following for the extractable energy

$$\frac{E_{\text{extractable},f}}{M_f} \approx 0.071 + 0.235(\chi_f - 0.686) + \mathcal{O}((\chi_f - 0.686)^2), \quad (4.3.2)$$

which would explain the almost one-to-one correspondence between final spin and extractable energy — as our simulations deviate very little from the 0.686 value and the linear approximation holds.

The extractable energy is found to vary in the range $E_{\text{extractable}} \in [0.064, 0.097]$, with the higher values corresponding to the most relativistic scenario ($p/M = 0.49$) because these are the ones with highest induced spins, which represent the BHs which we could extract most energy from.

Moreover, in Fig. 21 we see that the final mass increases for larger values θ/θ_0 , especially in the scenarios $p/M = 0.3675$ and $p/M = 0.49$; while for other cases there is a clearer change in the trend around the reference critical point, reaching an almost zero slope for the orange curve, while even reaching a negative slope for the blue one. It is not evident why this change in behavior occurs; the interplay of GW emission with the incidence angle (as explained earlier), as well as spin-induction, and horizon absorption effects, may be crucial to understand where these trends come from.

Further studies need to find proper correlations between the spin tendencies and the mass and irreducible mass — which seems to achieve a negative slope from the critical point forward for all scenarios on the right panel in Fig. 21 — tendencies, in order to comprehend what the exact interplay of angular momentum, energy, spin-induction, and mass increase from horizon absorption⁹ produces which produce the complex behaviors observed throughout this study of dynamical captures. We also intend to compare the computed values of final spin and final mass for each case with the predictions of quasicircular and head-on mergers with a given (effective) spin, expecting to find our simulations lying in between these other types of events.

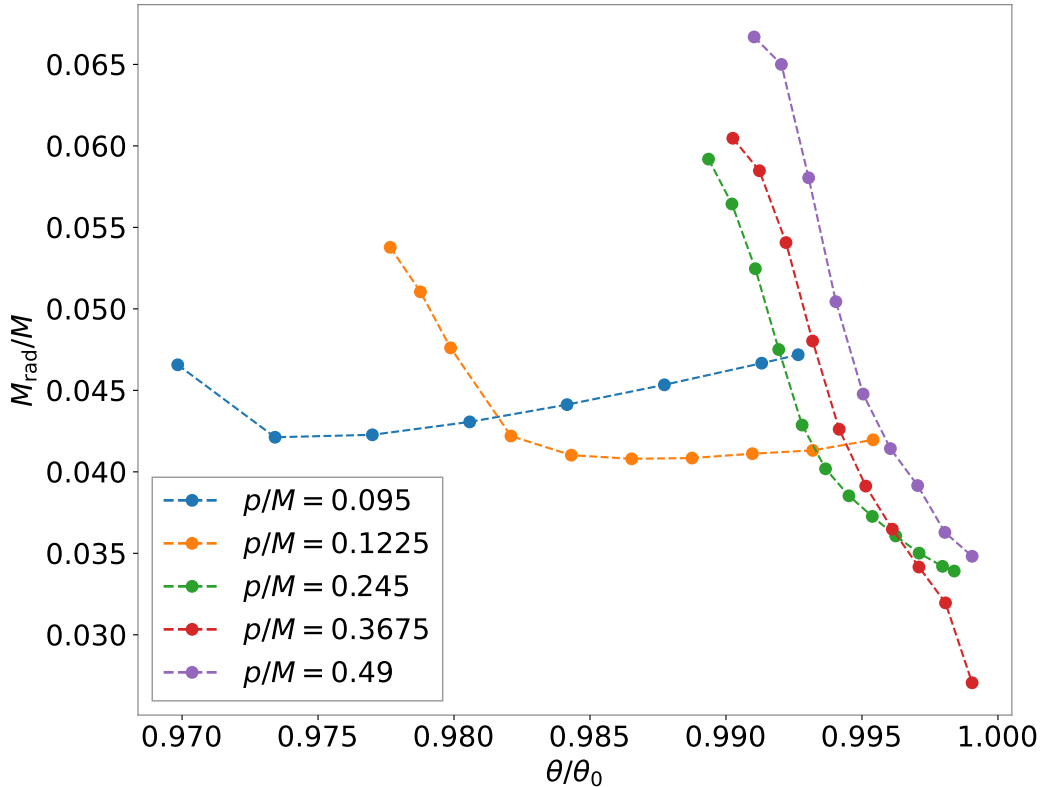


FIGURE 22: Radiated mass after the merger with respect to the normalized incidence angle θ/θ_0 for each scenario.

⁹Horizon absorption would explain how the M_f of some cases exceeds the value $1M$ (see the left panel on Fig. 21), because after the CHE we get masses up to $m/M \sim 0.52$, increasing the total ADM mass.

4.3.3. Analysis of radiated mass

Finally, we can also obtain the radiated mass during the merger. When two BHs fuse together, they emit vast amounts of energy and angular momentum. We already know that the remaining angular momentum transforms into spin. Notably as well, part of the emitted energy is actually mass of the initial system. In general, the mass of the final BH is smaller than that of the BHs right before the merger, and thus we can define the radiated mass of an equal-mass merger as

$$M_{\text{rad}} = 2m - M_{\text{f}}. \quad (4.3.3)$$

One subtlety that must be mentioned is that m is the mass of one of the BHs before the merger, and since we have horizon absorption, in general $2m \neq M$, where M is the total mass of the system at the start of the simulation. So, to account for the actual mass of the BHs before they fuse together, we compute the value (m) at an time of $30M$ before they merge.

We plot these results in Fig. 22, where we see that, in general, more head-on mergers imply more radiated mass; as well as the fact that more mass tends to be radiated for more relativistic encounters. We measure that these events emit 3% – 6% of their total mass in the form of GWs, which is a range within the typical values detected in LVK (as an example, the famous event GW150914 emitted around 4.6% of the total initial mass during the merger [1, 3]).

It is possible that in the near future, in our detectors we can find events where significant amounts of mass (for example 8% – 10%) was radiated away in a GW burst, which could mean that two highly relativistic BHs (with linear momenta $p/M \gg 0.49$) have merged with an almost-head-on trajectory, which could be used to check if it is a DC. In the near future, when we have enough cases to include the $p/M = 0.75$ scenario, we will check how much mass is radiated for these values of linear momenta.

5. Conclusion

In this MSc Thesis we have first (Chapter 1.) delved into the theory and general details of Numerical Relativity, more specifically describing the properties of our simulations. Going through the basics of gravitational-wave formalism, Kerr black holes and GR orbits; as well as describing some fundamental concepts of ADM formalism, which are necessary to build a proper NR simulation and comprehend some key aspects of this project.

Secondly, we introduced Chapter 2., in which we described the ETK, which was the software used to compute all the simulations. We also described the processing and post-processing tools we use to obtain relevant outputs that have been shown throughout the entirety of the next chapters 3. and 4..

Moreover, in this second chapter we used examples for both pure CHEs and DCs (Secs. 2.3. and 2.4., respectively) to observe some key aspects of these simulations such as the evolution of the BH spins, which suffer prominent spin-induction, that then lead to a merger with high spins ($\chi_f \sim 0.74$), in the case of DCs. Another key aspect was the mass increase due to horizon absorption of GWs, as described by [23], which in our case, reached a maximum value of $\sim 5\%$ for the most relativistic scenario. We also introduced some Hamiltonian formalism to explain why spin-induction is produced.

After that, in the third chapter (3.), we have investigated the phenomenon of spin induction that occurs in both black holes during a close hyperbolic encounter. Our study is based on numerical relativity (NR) simulations, following the methodologies of Refs. [47, 59], and includes the introduction of initial spins for the black holes. The more detailed version of this study is described in our publication [72].

We first examined a set of simulations where both black holes were initialized with equal spins aligned with the orbital angular momentum. In this scenario, we demonstrated that the magnitude of the spin-up effect diminishes as the initial spin increases. This behavior is attributed to spin-orbit coupling, as described in Eq. (2.5.2), and is well captured by an exponentially decaying function, as illustrated in Fig. 10.

Next, we explored more general configurations with unequal initial spins, including both aligned and orthogonal orientations relative to the orbital angular momentum. For a fixed initial effective spin, we found that the resulting induced effective spin increases with larger values of $\chi_1(0)$ and $\chi_2(0)$. This dependence is well modeled by a second-order polynomial for both the aligned cases and for the z -component in the orthogonal configurations, as shown in Figs. 11 and 13. Unlike the equal-spin case, these configurations exhibit a non-zero weighted spin difference $\delta\chi$, which follows an approximately linear trend (see Fig. 11). By combining the observed trends, we provide effective quadratic expressions for $\chi_1^z - \chi_1^z(0)$ and $\chi_2^z - \chi_2^z(0)$, as presented in Fig. 12. An intuitive depiction of this effect is provided in Fig. 8.

Furthermore, we analyzed the spin-up in the x and y components resulting from spin precession, described by Eq. (3.2.9). These components also exhibit an approximately linear behavior, as displayed in Fig. 13.

Our results show that introducing initial spin significantly influences the black holes' trajectories — particularly the scattering angles — in close hyperbolic encounters (CHEs), see Fig. 15.

These findings are relevant for analytical modeling of spin effects, as pursued in [23], extending the results from [47].

A precise understanding of the spin induction effect is crucial for studying the clustering behavior of black holes, especially in scenarios involving primordial black holes. CHEs occurring in dense black hole environments are expected to play a significant role in shaping the spin distribution of the resulting population [40]. This highlights the necessity of going beyond spinless scattering studies [47, 59], by including configurations with both parallel and orthogonal spins relative to $\vec{L}$, as accomplished in this work.

Our NR simulations contribute to the broader understanding of spin induction, offering diverse perspectives as various regions of parameter space are investigated. Future research should extend this study to include spinning black holes with unequal masses. Ultimately, a comprehensive description of spin induction will require support from analytical methods, such as post-Newtonian, post-Minkowskian, and effective one-body approaches. In the meantime, NR simulations remain a vital and powerful tool for probing the rich phenomenology of this effect.

Finally, in Chapter 4. we carried out a first approach to the analysis of dynamical captures leading to a merger. Here, we have tried to model the waveform of such events and studied the trends of the spins, masses and irreducible masses of the final black holes with respect to the (“normalized”) incidence angle.

With the explained procedure for waveform-fitting, we obtained a very accurate fit of the distance between the CHE and merger-ringdown emissions ($\Delta t/M$, Eq. (17)) which provides a characterization of the threshold angle (θ_0) that separates purely scattering events from DCs.

Moreover, we have seen in Subsec. 4.2.1. that the *sin-Gaussian* parametrization described by Eq. (4.2.1) models the Weyl scalar extremely well. The table of parameters shows quite good precision in the estimated trends for these parameters with respect to θ/θ_0 .

On the other hand, the merger and ringdown parameters do not show evident trends (as seen in the Annex), even having oscillatory behavior in the estimated phases, having values clustered around the same orders of magnitude for most parameters, complicating the analysis of the model described in Subsec. 4.2.2. by Eqs. (4.2.1) and (4.2.2). Nevertheless, this model shows that, as a first approach, the agreement with the NR Weyl scalar is quite good (see Fig. 16); and provides a good initial description of the frequencies, amplitudes, etc. that contribute to this final emission.

As a final note for this chapter, we showed the Figs. 20, 21 and 22 which highlight the dependencies of the final BH spin and extractable energy; ADM mass and irreducible mass; and radiated mass, respectively, in terms of θ/θ_0 for each scenario. We stress the need for a proper understanding of the trends observed here, even though we have stated that the interplay of GW emission — depending on how close we are to a head-on collision — and initial angular momentum, as well as the spin induction and other relativistic mechanisms (such as horizon absorption) are crucial to fully comprehend the behavior of these plots.

In conclusion, this MSc Thesis paves the way towards a proper waveform model of DCs, that could as well describe their corresponding effects in the orbital dynamics, spin-inducing and mass absorption mechanisms; and the intrinsic properties of the initial and final BHs. In the near future, we will continue expanding this work, leading to more unprecedented results on the events of CHEs and DCs. The results of these works are expected to gain relevance as detector sensitivity improves, allowing us to detect signals coming from such events.

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Annex

A. Fitted values for the estimated parameters of (CHE) Ψ_4

TABLE 4: Calculated fitting values with respect to θ/θ_0 for the estimated parameters of the CHE portion of the Weyl scalar according to the methodology described in Sec. 4.2.1.. In particular, the “Linear” and “Exponential” fits are shown in Eqs. (4.2.5) and (4.2.6). Notice, as well that as mentioned in the respective subsection, we have imposed both the imaginary and real parts to have the same frequency, while leaving the phases as free parameters. The fits for these last variables are sufficiently good because they fulfill the requirement of $\pi/2$ of phase difference, up to a certain numerical error (compare the c parameter for $\phi_{R,I}$). We show the type of fit for each parameter, and the respective values and uncertainties on the last digits(s) at 1σ in parentheses.

CHE Parameter	Fit	p/M	a	b	c
A	Linear	0.095	−0.601(10)	0.648(9)	—
		0.1225	−0.65(2)	0.79(2)	—
		0.245	0.29(20)	0.18(20)	—
		0.3675	0.84(17)	−0.61(17)	—
		0.49	1.57(20)	−1.30(20)	—
σ	Exponential	0.095	0.0342(130)	162(12)	10.8(9)
		0.1225	0.0810(134)	192(20)	10.0(5)
		0.245	0.0819(210)	370(22)	8.73(8)
		0.3675	0.353(44)	226(10)	8.34(7)
		0.49	0.370(65)	238(16)	9.13(10)
t_0	Linear	0.095	−255(28)	543(29)	—
		0.1225	−384(31)	621(30)	—
		0.245	−382(34)	531(33)	—
		0.3675	−302(19)	426(19)	—
		0.49	−313(17)	429(18)	—
ω	Linear	0.095	−1.065(17)	1.338(17)	—
		0.1225	−0.874(43)	1.192(42)	—
		0.245	0.051(119)	0.390(120)	—
		0.3675	−0.420(120)	0.915(121)	—
		0.49	−0.554(83)	1.064(82)	—
ϕ_R	Exponential	0.095	0.228(56)	82.94(748)	−5.88(12)
		0.1225	0.153(39)	132.7(102)	−5.35(7)
		0.245	0.208(45)	214.9(182)	−4.00(6)
		0.3675	0.519(48)	140.4(68)	−3.16(5)
		0.49	0.555(70)	142.5(97)	−2.00(8)
ϕ_I	Exponential	0.095	0.227(57)	83.00(748)	−4.30(11)
		0.1225	0.153(39)	132.6(100)	−3.78(8)
		0.245	0.205(41)	215.7(183)	−2.45(6)
		0.3675	0.500(48)	142.6(71)	−1.55(5)
		0.49	0.628(80)	134.0(95)	−0.516(90)

B. Figures of estimated parameters for the Merger and Ringdown

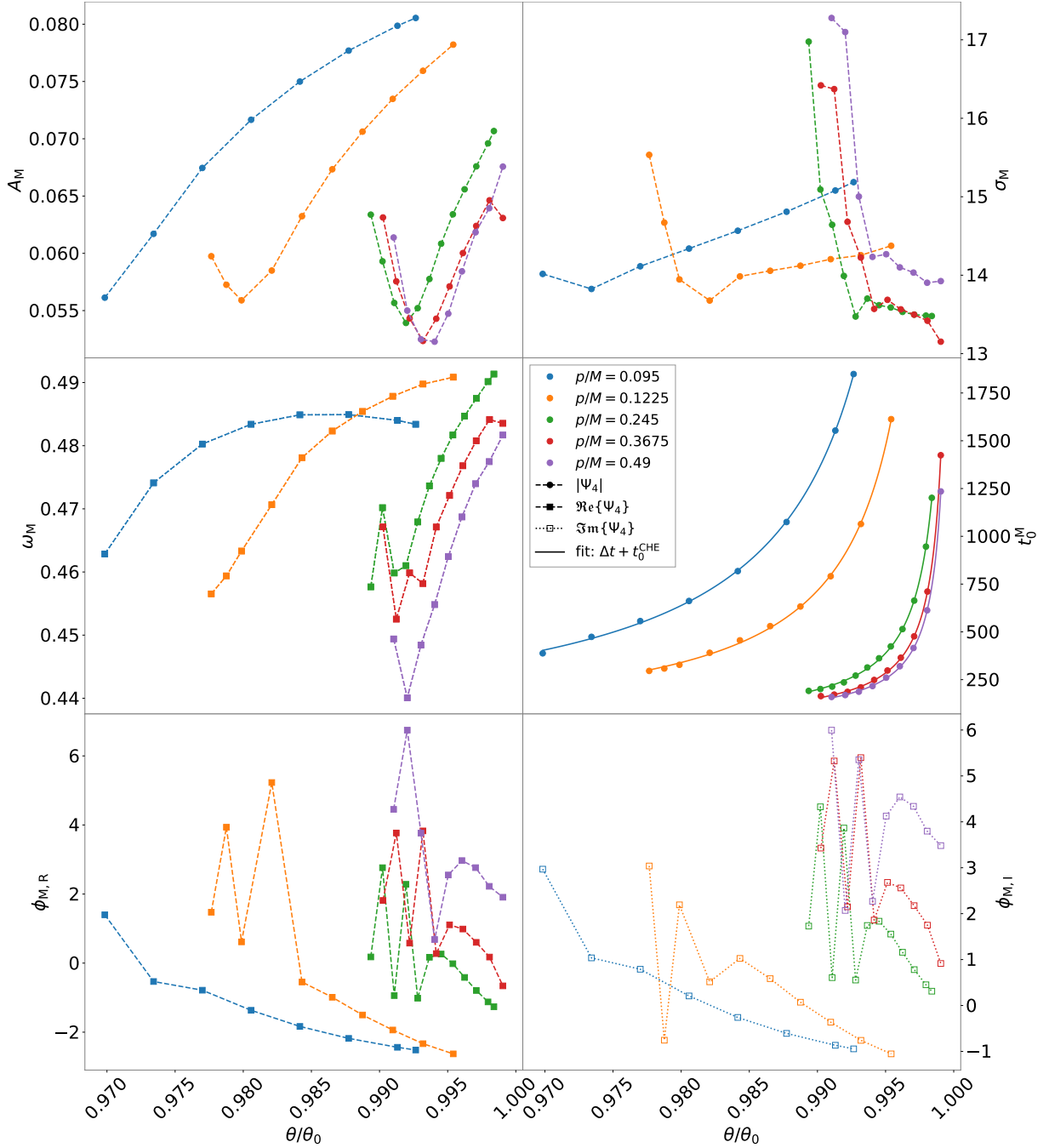


FIGURE 23: Estimated parameters from the fitted merger portions of the Weyl scalars for all values of p/M . We also show how we compute these values: the ones represented with a solid dot are calculated by fitting the modulus of the Weyl scalar, while the ones with solid and empty squares are computed by fitting the real and imaginary parts, respectively. The fitted frequencies overlap for both components because we have imposed then to find a common best-fit.

In this section of the Annex we show the obtained values for each case for its merger emission (Fig. 23). The procedure is explained in full detail in Subsec. 4.2.2.. We can see that general trends can be seen (specially for t_0^M , A_M and ω_M), but there is room for improvement. This is especially evident for the phases, more specifically, for low θ/θ_0 in each scenario, where oscillations appear and no clear trends can be deduced. A very similar phenomenon occurs for the estimated

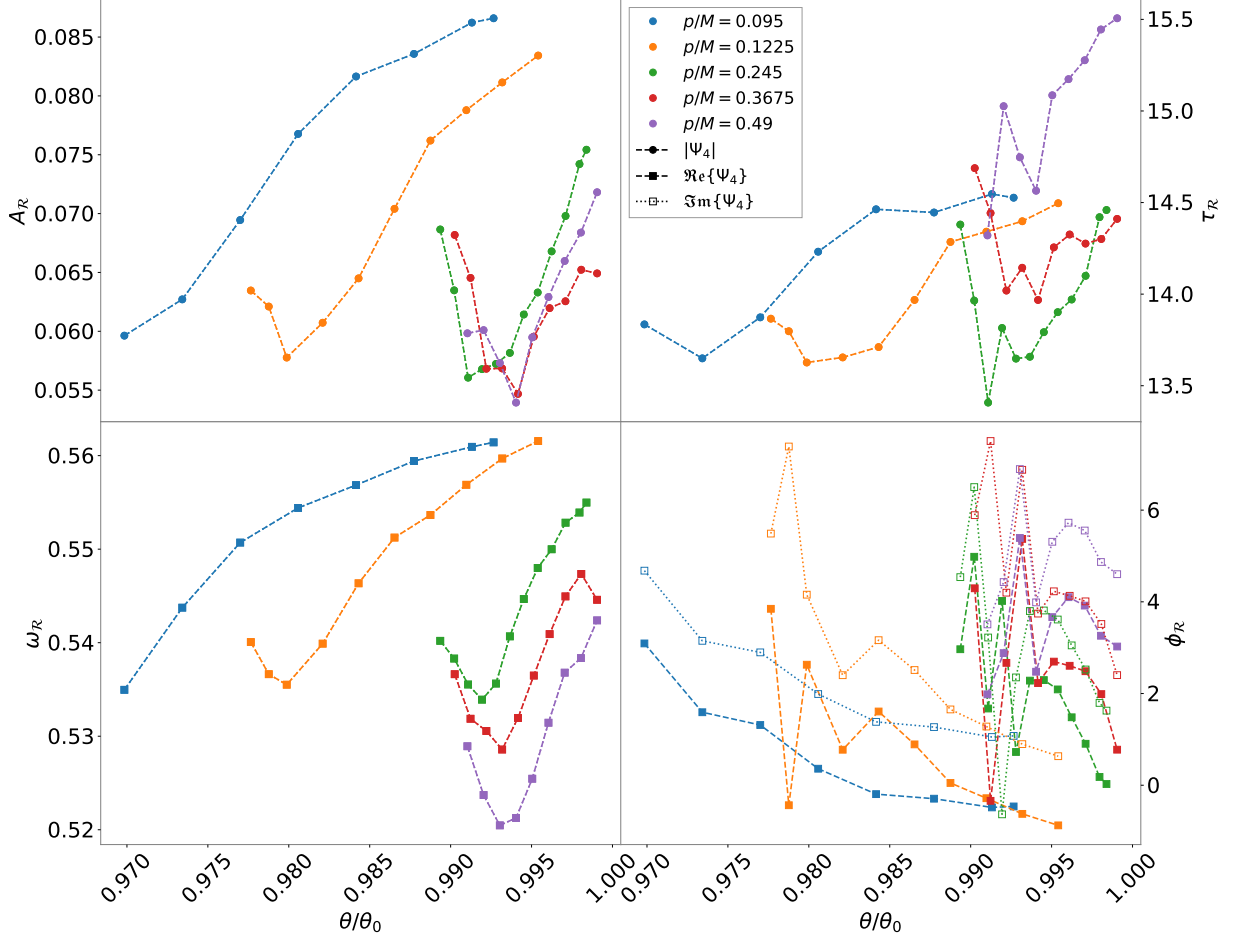


FIGURE 24: Estimated parameters of the first mode of the ringdown portion of the Weyl scalars for all values of p/M . We also show how we compute these values: the ones represented with a solid dot are calculated by fitting the real part of the Weyl scalar, while the ones with and empty dot are computed by fitting the imaginary part, respectively. The fitted frequencies overlap for both components because we have imposed then to find a common best-fit.

ringdown parameters (Fig. 24), where oscillations come at play once again.

As no clear trend have been found yet, we have not carried out any fits to the parameters as done for the CHE parameters, but we will attempt to fix the “oscillation issues” in the near future, to obtain clearer trends, possibly allowing us to fit the different parameters with respect to θ/θ_0 (maybe even using the same expressions utilized for the CHE, Eqs. (4.2.5) and. 4.2.6). Nevertheless, this study helps us see the orders of magnitude that the GW emissions in DC events have, paving the way towards a future DC waveform model.

C. An example of the overlap of the tails of both emissions

In Fig. 25 we show the performance of the combination of fits described in Sec. 4.2. for a case where the tails of both *sin-Gaussian* distributions combine with each other. This is the main reason we observe the oscillations (especially in the phases) in the Figs. 23 and 24.

As already mentioned, we will try to better describe these cases where the emission tails fuse with each other, maybe using a time dependent frequency that smoothly goes from the one

predicted for the CHE to the one estimated for the merger. That is reason for future work that will be done in the months that ensue the presentation of this Thesis.

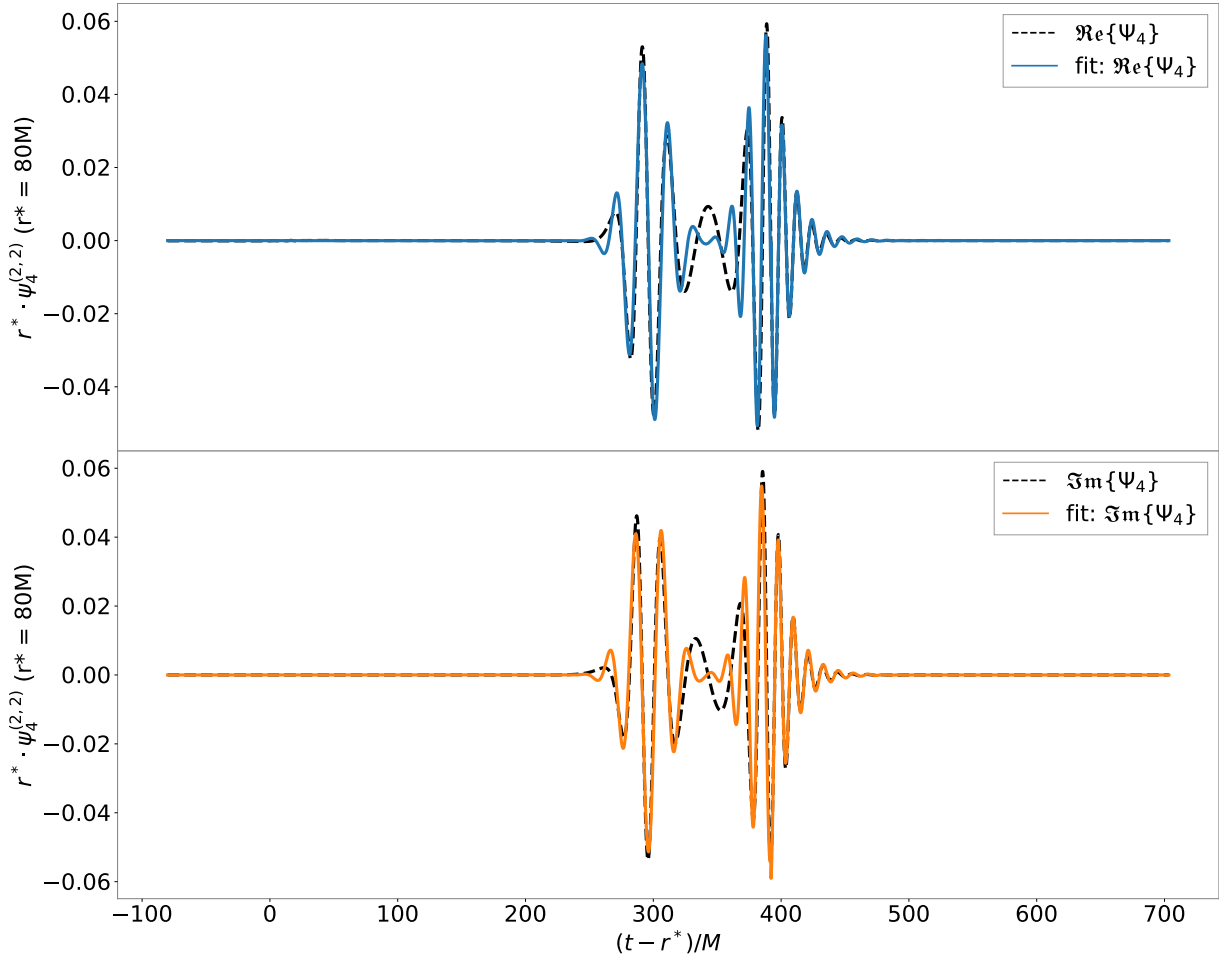


FIGURE 25: Fits of the real (blue) and imaginary (orange) parts — upper and lower panel respectively — along with the simulated signal (black, dashed) of the rescaled Weyl scalar at a detector at a distance $r^* = 80M$ from the CM. The initial conditions are $p/M = 0.095$ and $\theta = 6.2108^\circ$ (and, of course, zero initial spins).